failure1 — Native-IMU reconstruction: method and interpretation

Separate experiment: existing mocap identification and outputs are unchanged.

Identification uses native gyro / specific force with fitted thrust, gyro and accel first-order lags.

Orientation: gyro prediction + acceleration update recur; magnetic factors affect output only.

Only initial orientation / position and initial velocity use mocap. Later mocap is read-only evaluation.

Comparison uses original mocap timestamps inside IMU support; reconstructed attitudes are Slerped.

Native timestamps and firmware first-or-changed magnetic freshness proxy are retained.

Static sensor covariance = trace(unbiased static covariance)/3 times identity; no bias subtraction.

Offline local polynomial derivatives and their overlap covariance are included in the lag fit.

Uncertainty: Bingham-derived right-angle covariance and deterministic local Gaussian propagation.

Shared acceleration/attitude information uses declared conservative covariance bounds.

These algebraic bounds do not guarantee coverage of systematic acceleration or magnetic bias.

Magnetic output can be worse on these data; no mocap-based gating, tuning or recursive feedback.

Model reconstruction adds the native-IMU residual wrench time series to the physical dynamics.

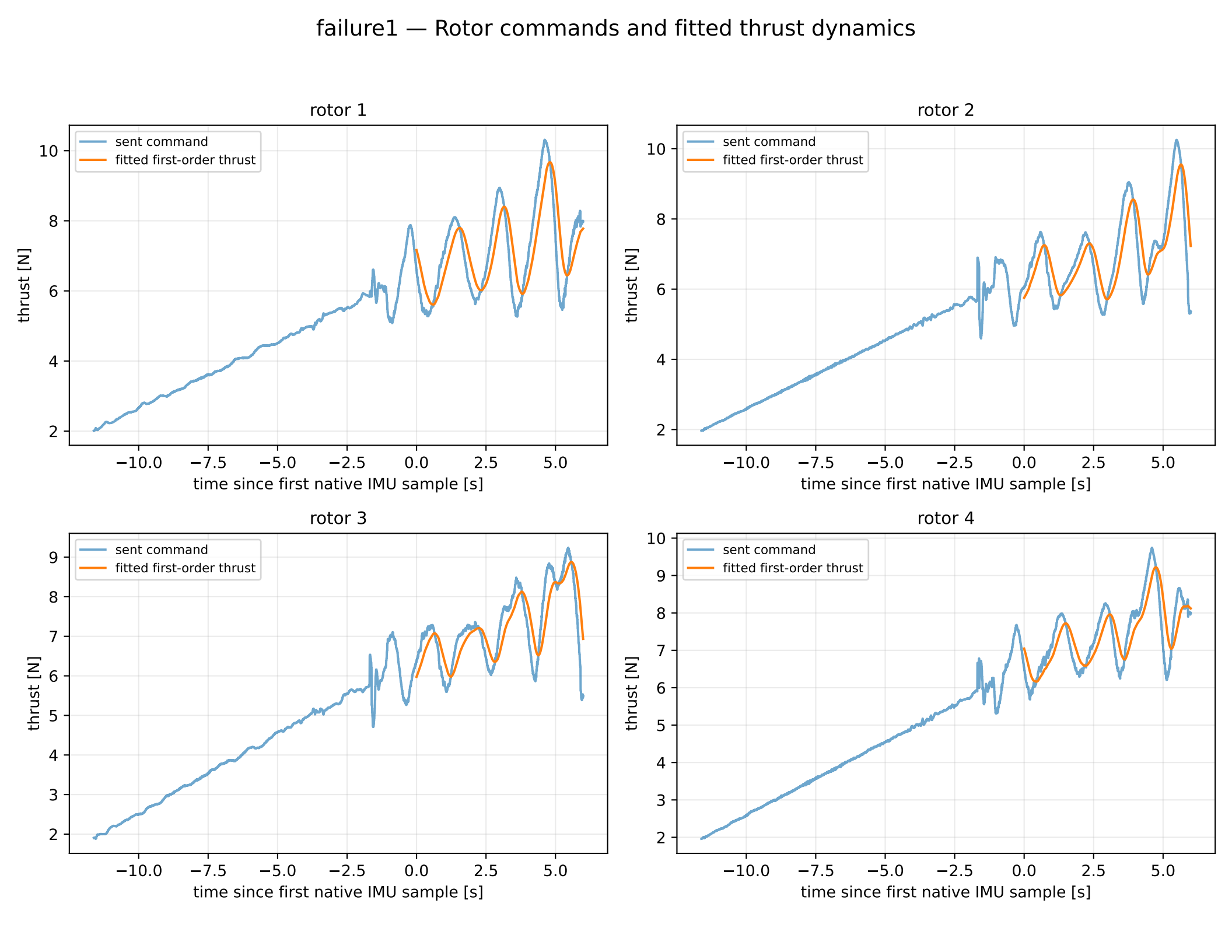
No Monte Carlo, artificial display variance, eigenvalue clipping or uncertainty-based sample selection.

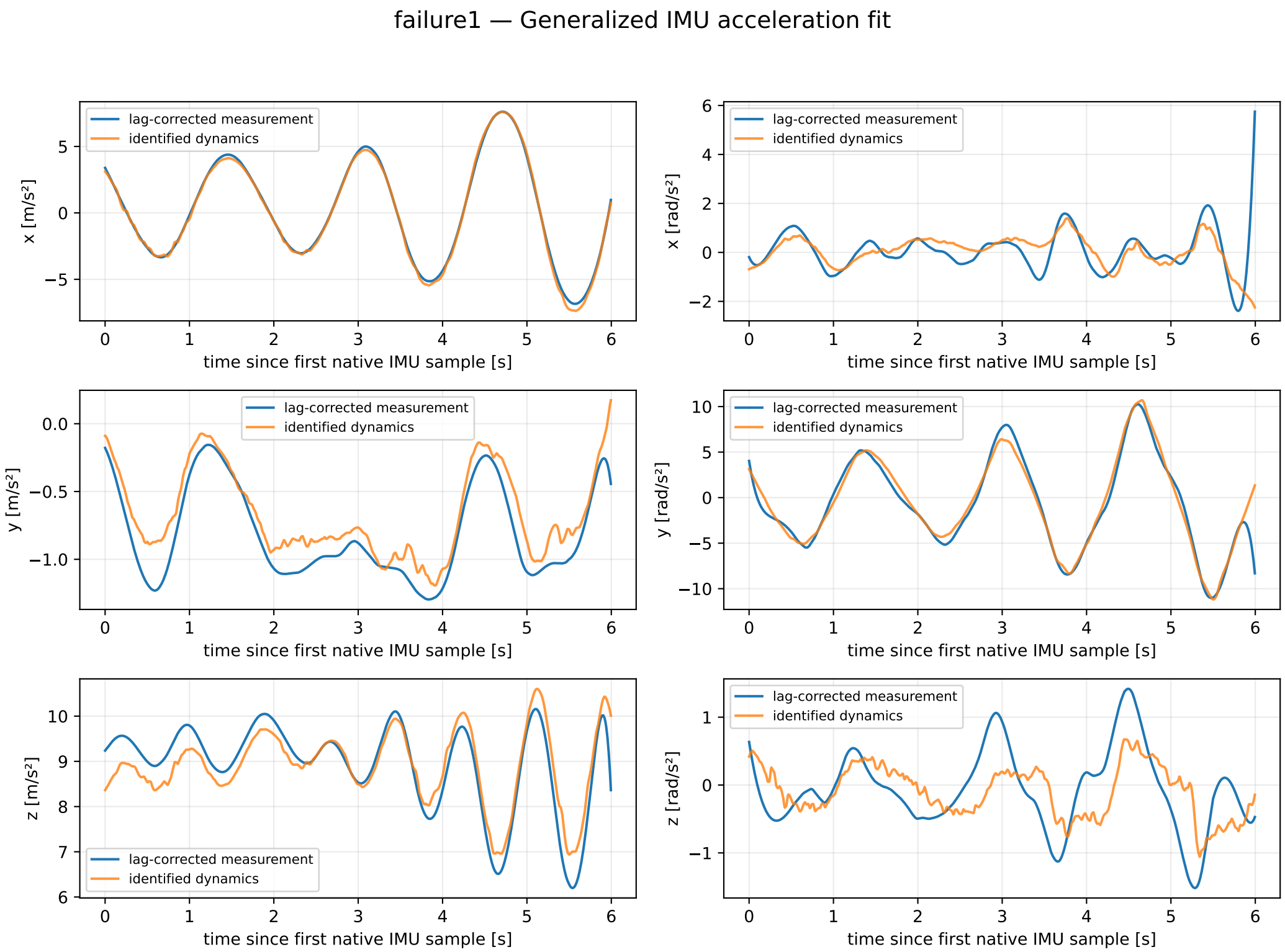
Mass / inertia / thrust scale use a common-scale gauge; chip lever arm is fixed, not calibrated.

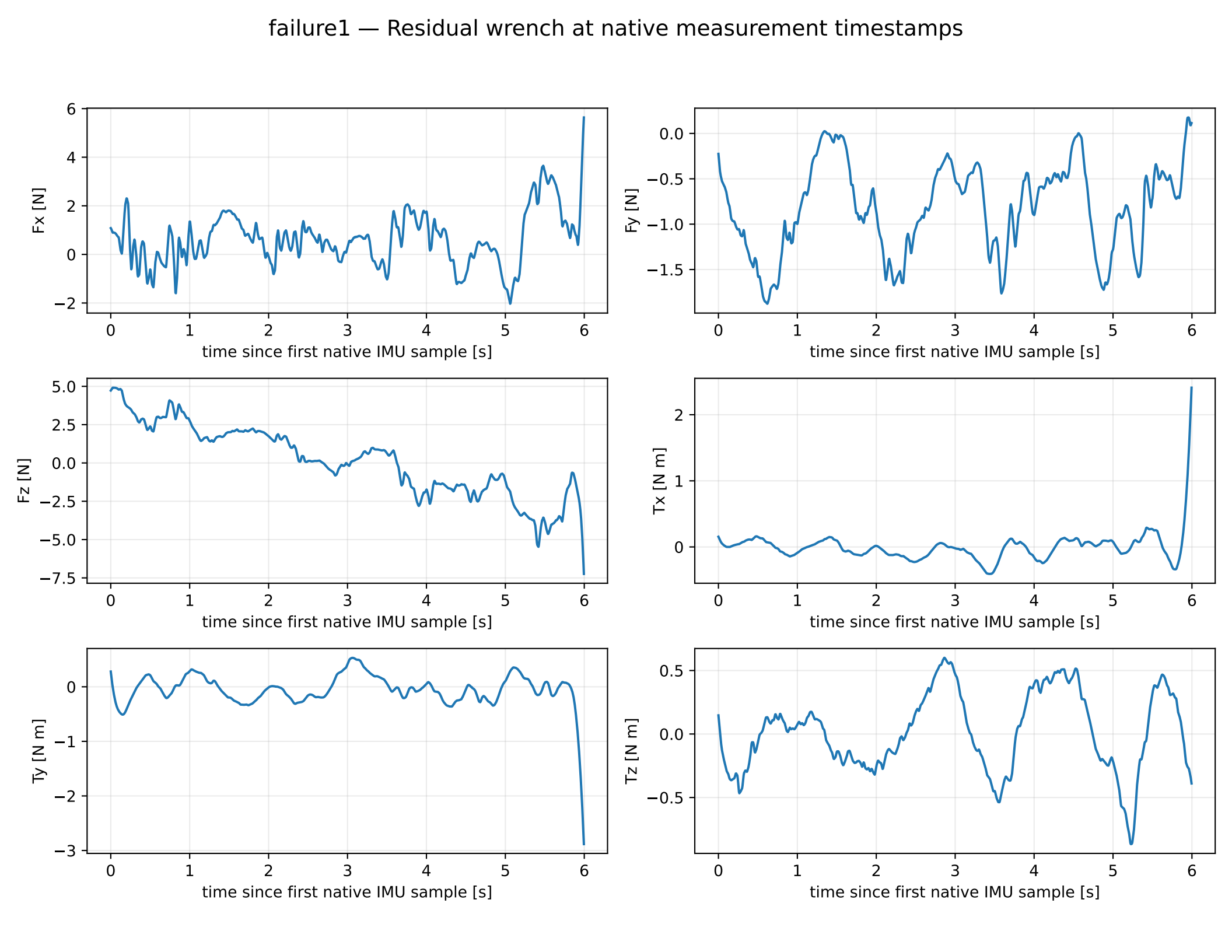
All 17 actual SI estimates and the complete joint covariance are appended at the end.

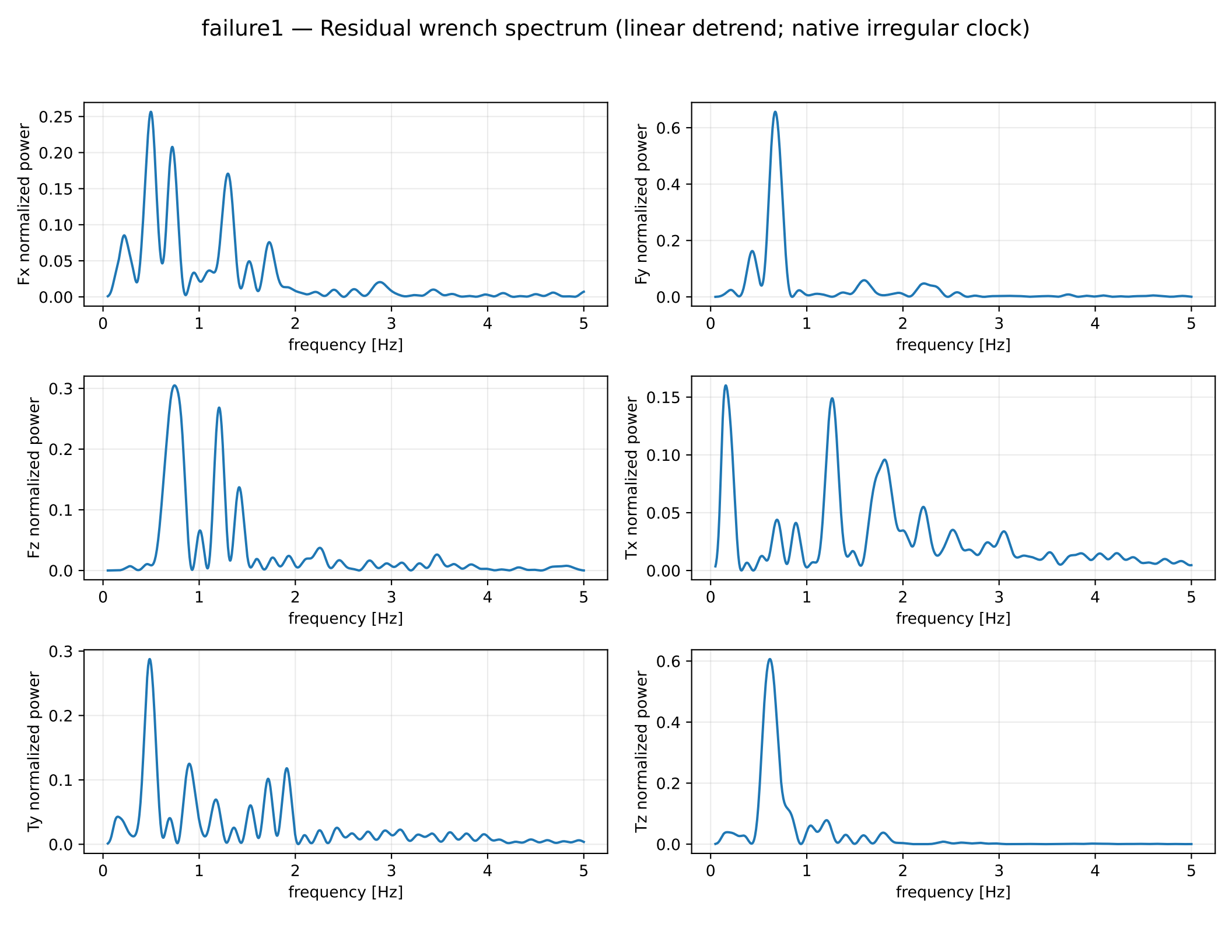
Native IMU support: 19.002181 to 24.997109 bag seconds; N = 1200.

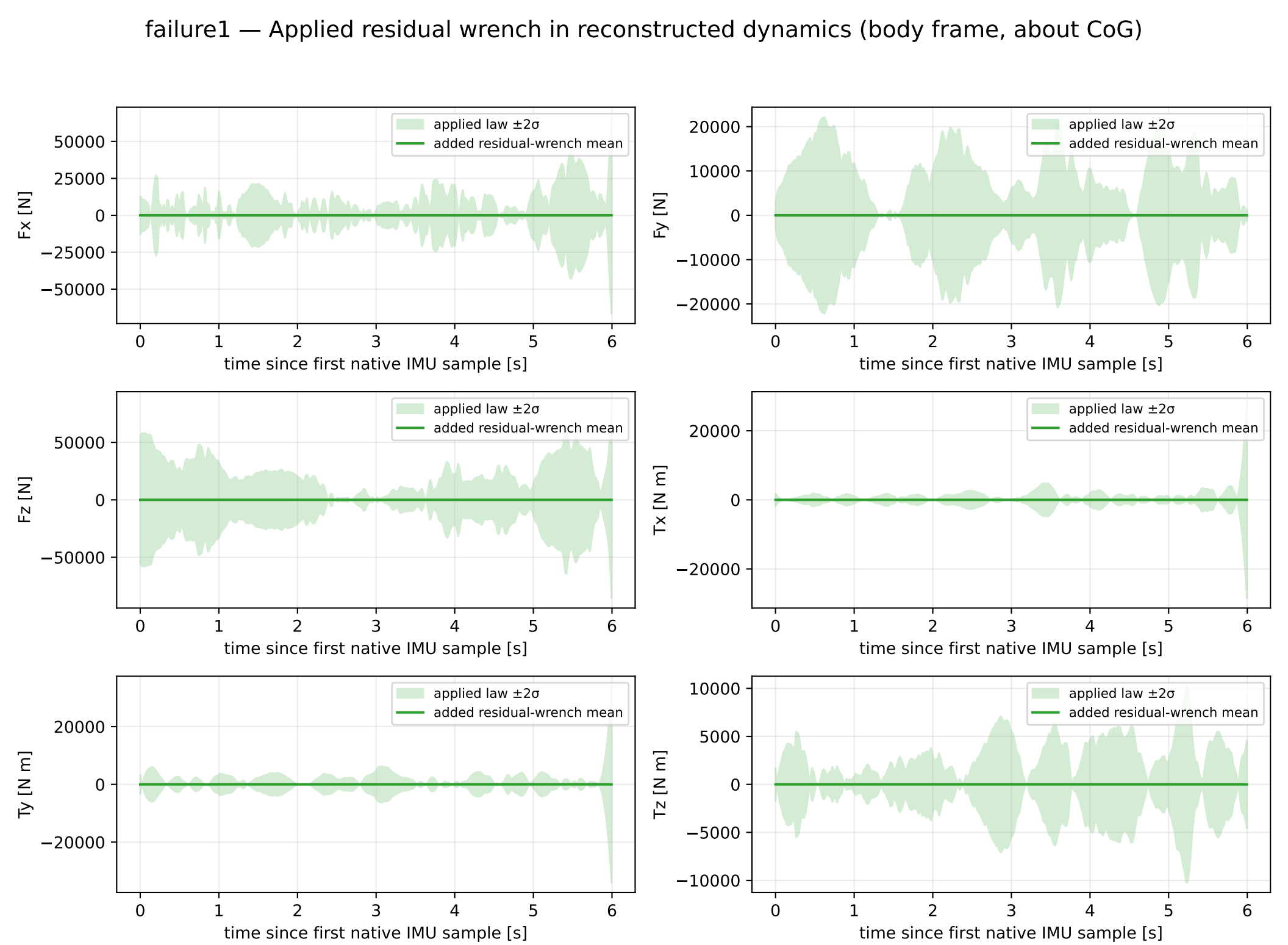
Individual PDF appendices and full-precision covariance CSV contain the same distribution.

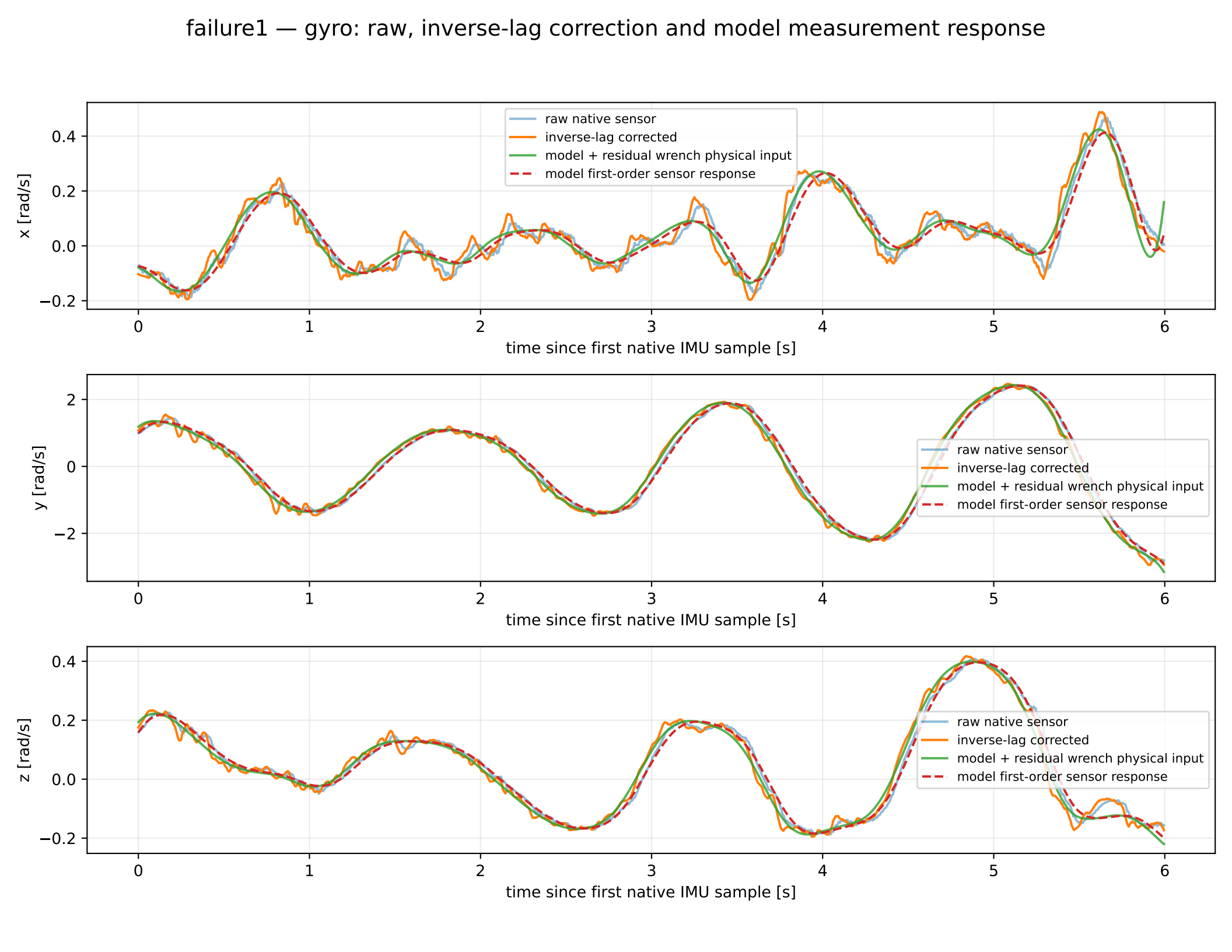

failure1 — Rotor commands and fitted thrust dynamics
rotor 1
rotor 2
rotor 3
rotor 4
sent command
fitted first-order thrust
thrust [N]
time since first native IMU sample [s]
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0

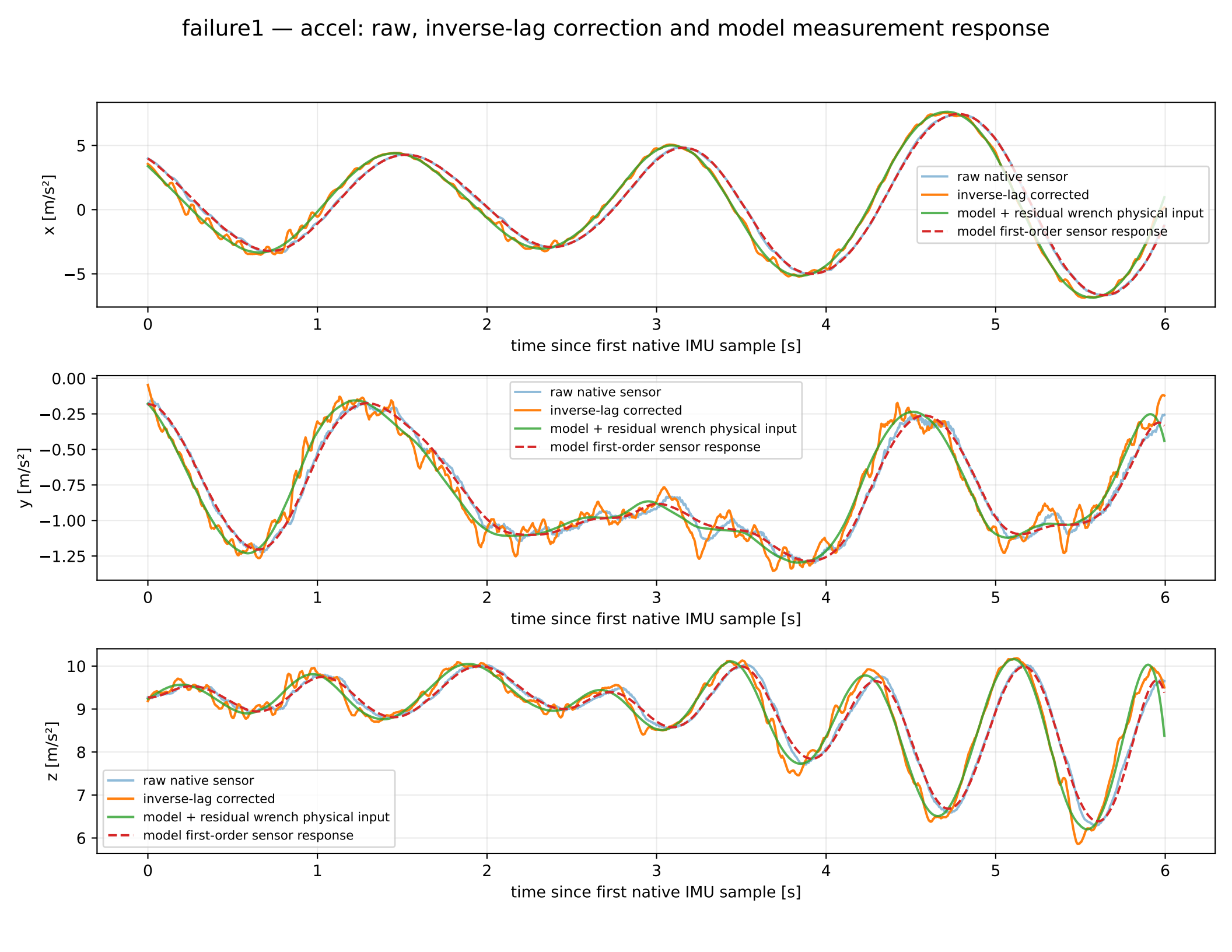

failure1 — Generalized IMU acceleration fit
lag-corrected measurement
identified dynamics
x [m/s²]
x [rad/s²]
y [m/s²]
y [rad/s²]
z [m/s²]
z [rad/s²]
time since first native IMU sample [s]

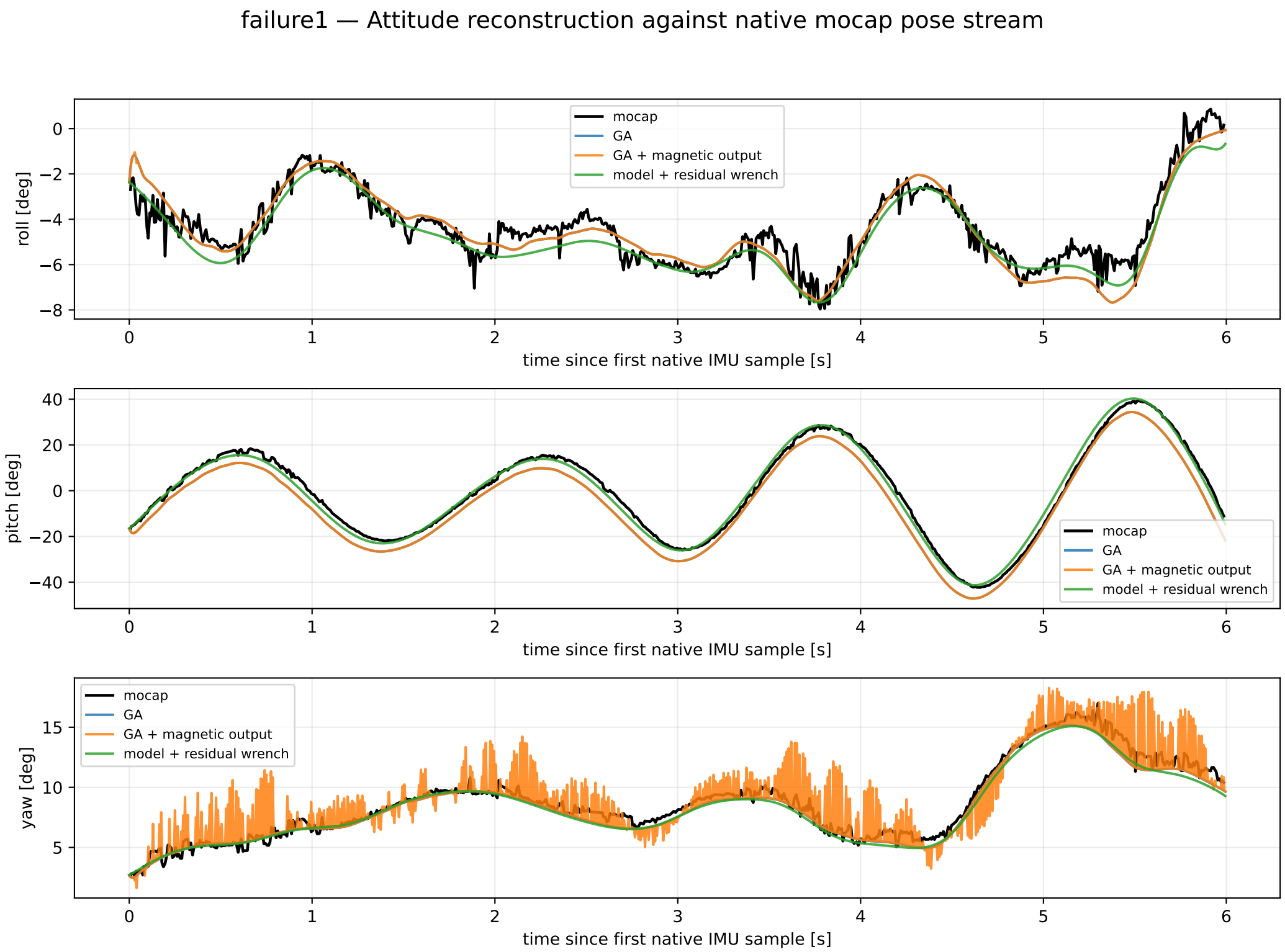

failure1 — Residual wrench at native measurement timestamps
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
time since first native IMU sample [s]

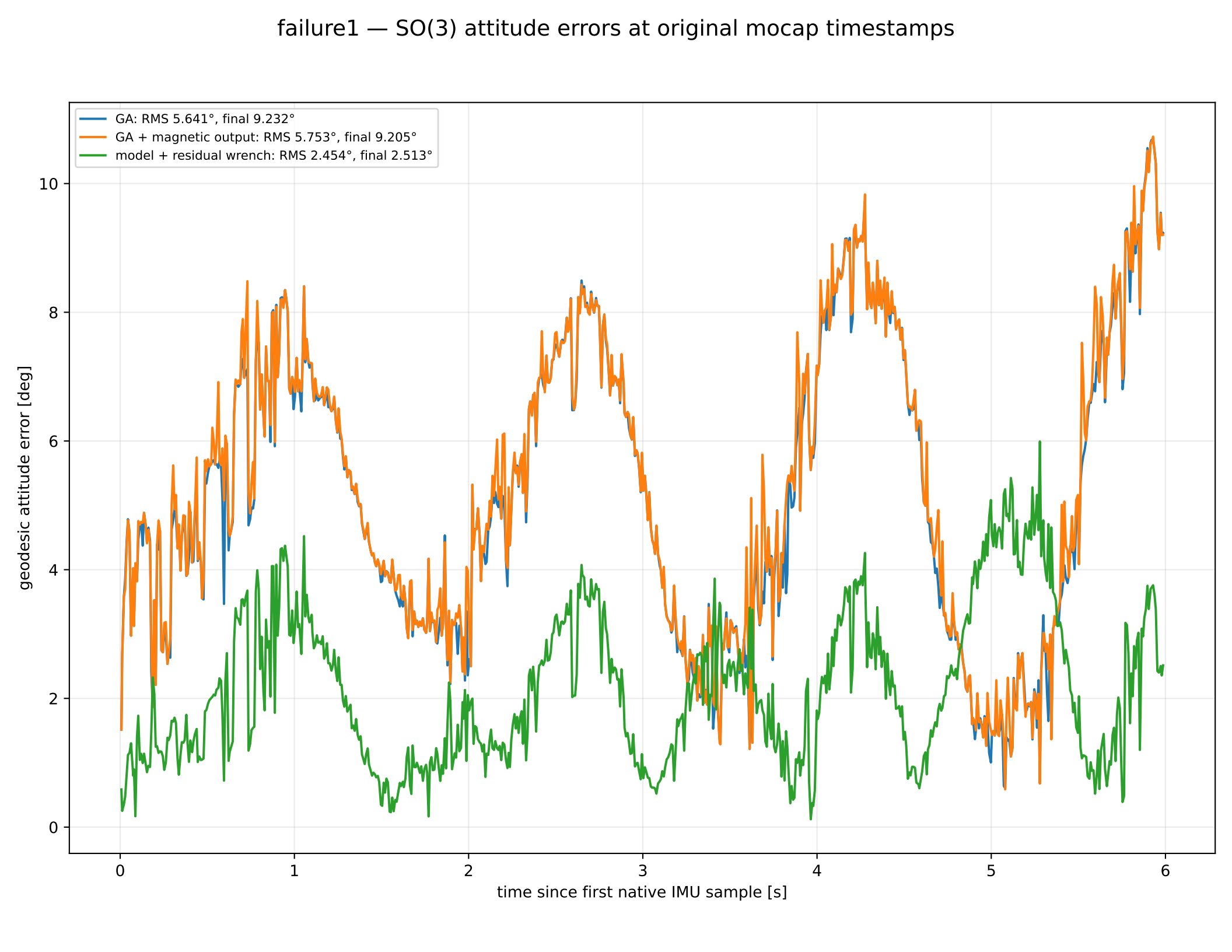

failure1 — Residual wrench spectrum (linear detrend; native irregular clock)
Fx normalized power
Fy normalized power
Fz normalized power
Tx normalized power
Ty normalized power
Tz normalized power
frequency [Hz]

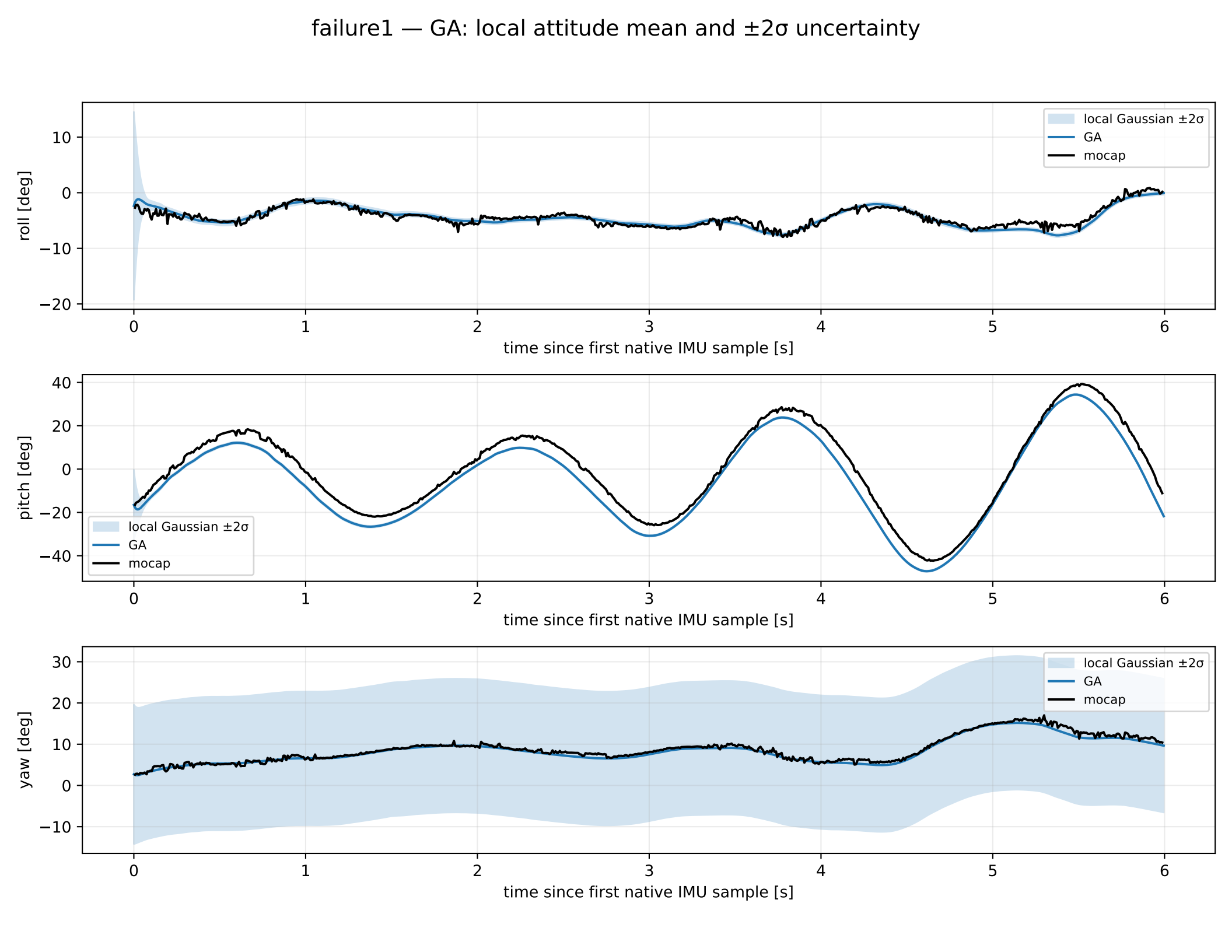

failure1 — Applied residual wrench in reconstructed dynamics (body frame, about CoG)
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
applied law ±2σ
added residual-wrench mean
time since first native IMU sample [s]
The native-IMU residual wrench is added to the dynamics; this is not a model-only free replay.

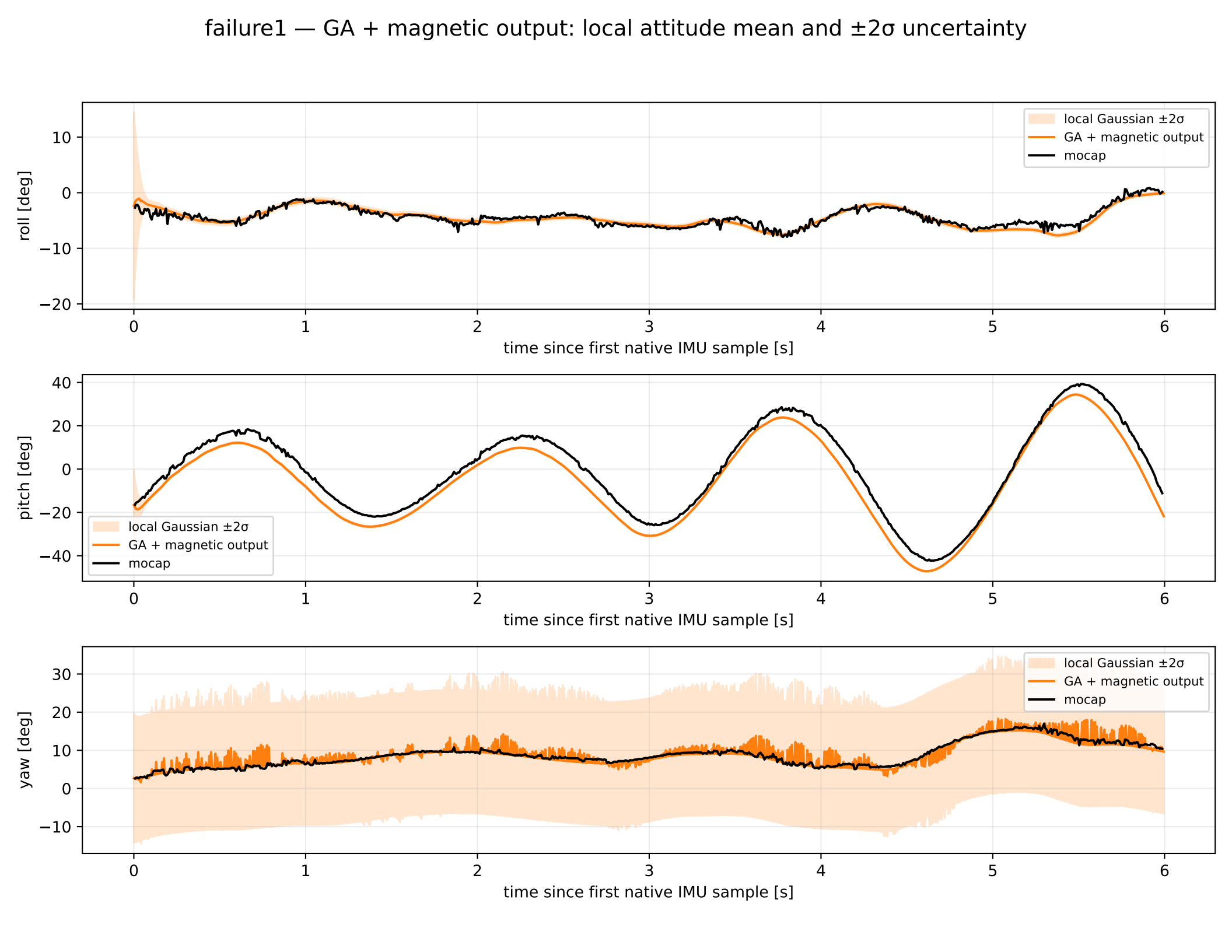

failure1 — gyro: raw, inverse-lag correction and model measurement response
raw native sensor
inverse-lag corrected
model + residual wrench physical input
model first-order sensor response
x [rad/s]
y [rad/s]
z [rad/s]
time since first native IMU sample [s]

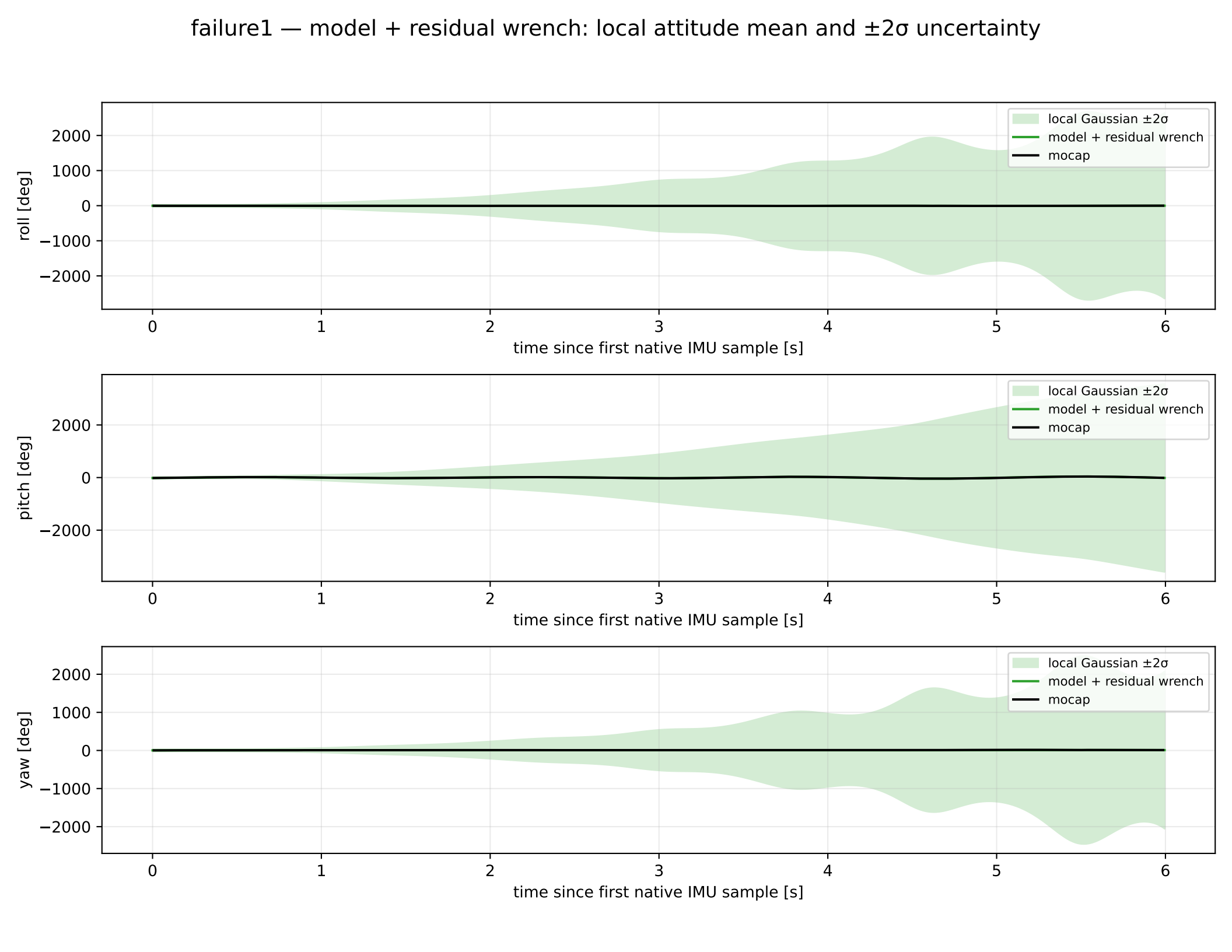

failure1 — accel: raw, inverse-lag correction and model measurement response
x [m/s²]
y [m/s²]
z [m/s²]
time since first native IMU sample [s]
raw native sensor
inverse-lag corrected
model + residual wrench physical input
model first-order sensor response


failure1 — Attitude reconstruction against native mocap pose stream
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
mocap
GA
GA + magnetic output
model + residual wrench

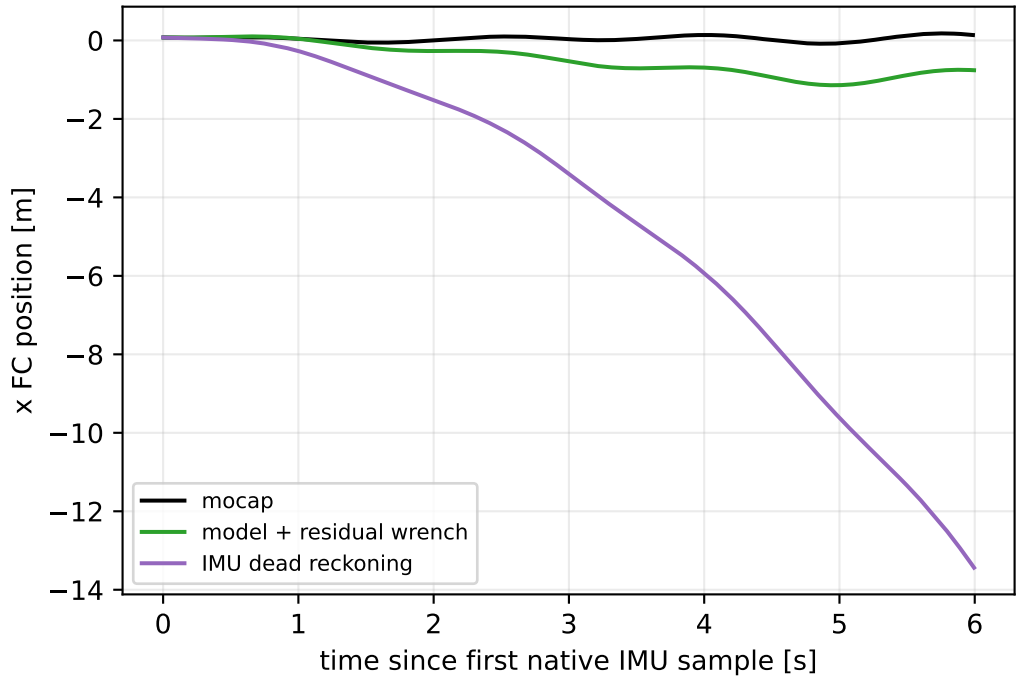

failure1 — SO(3) attitude errors at original mocap timestamps
GA: RMS 5.641°, final 9.232°
GA + magnetic output: RMS 5.753°, final 9.205°
model + residual wrench: RMS 2.454°, final 2.513°
geodesic attitude error [deg]
time since first native IMU sample [s]
0
2
4
6
8
10
0
1
2
3
4
5
6

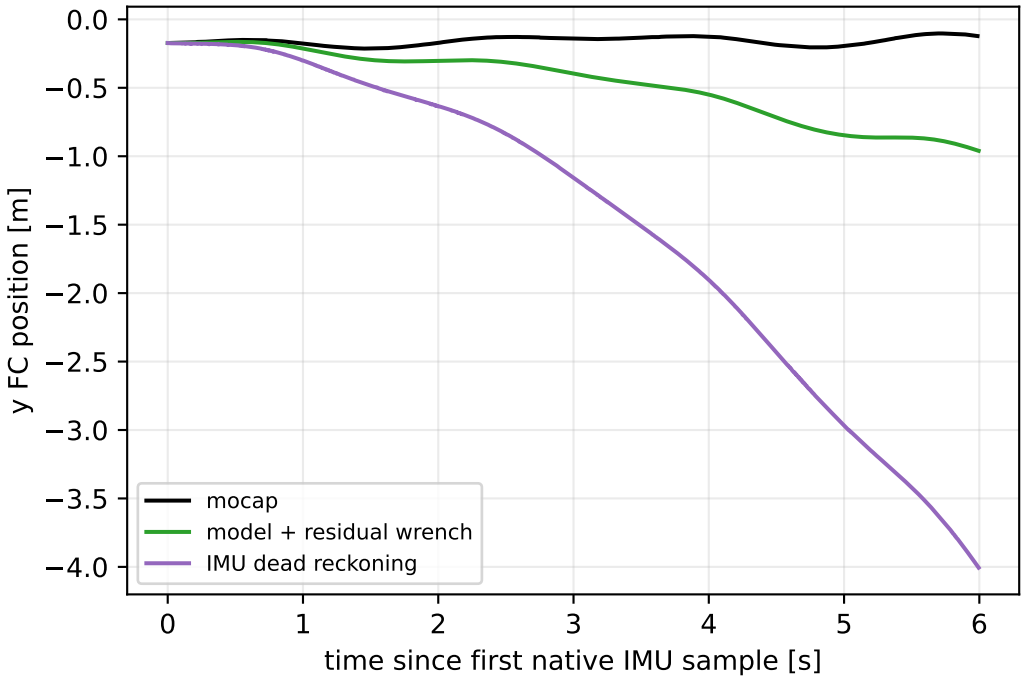

failure1 — GA: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA
mocap

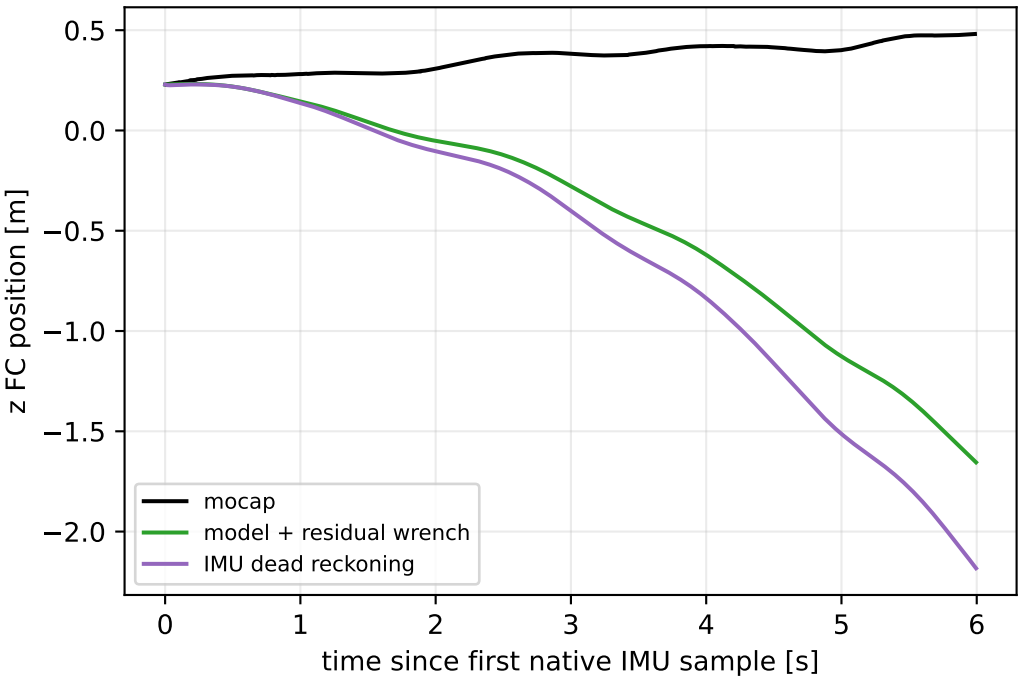

failure1 — GA + magnetic output: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA + magnetic output
mocap

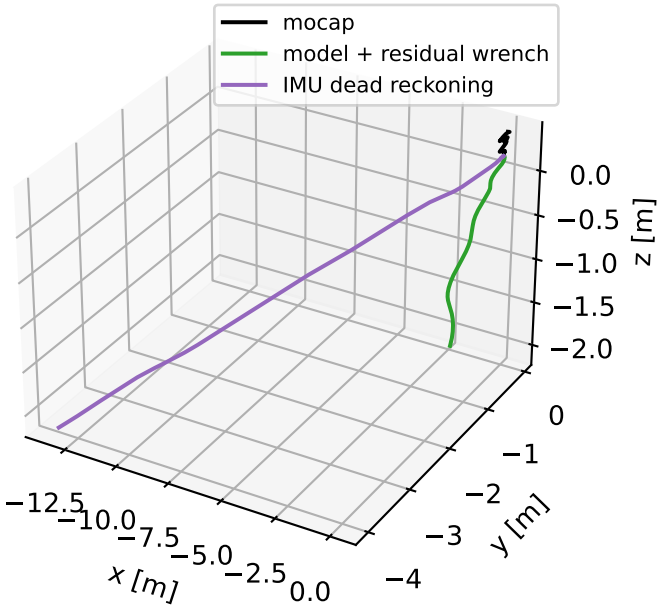

failure1 — model + residual wrench: local attitude mean and ±2σ uncertainty
local Gaussian ±2σ
model + residual wrench
mocap
roll [deg]
pitch [deg]
yaw [deg]
2000
1000
0
−1000
−2000
0
1
2
3
4
5
6
time since first native IMU sample [s]

failure1 — Position reconstruction (mean trajectories)



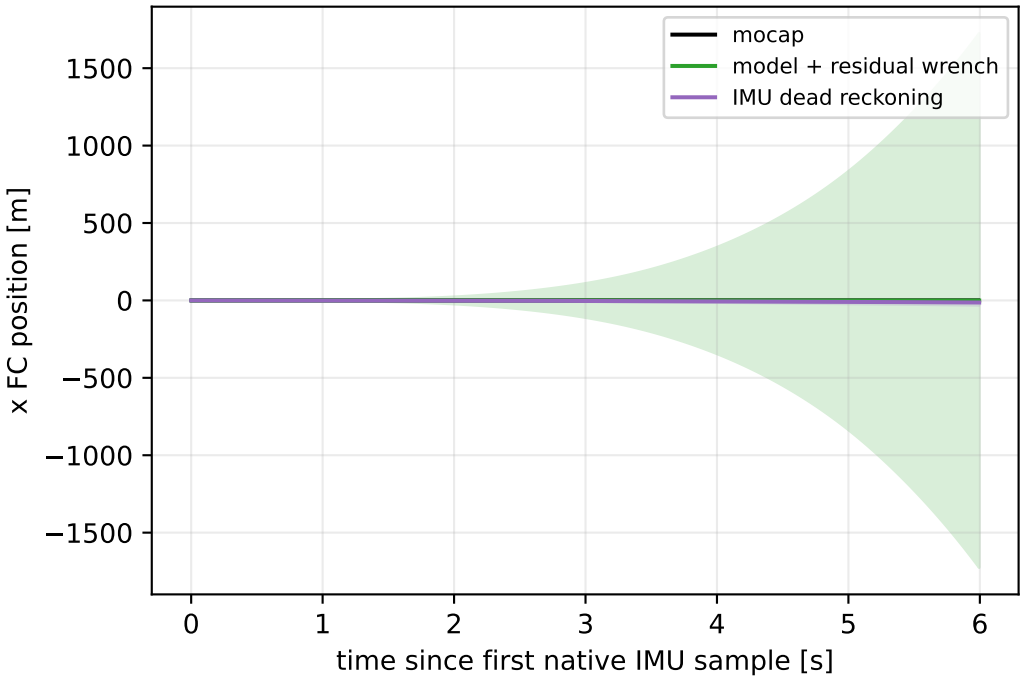


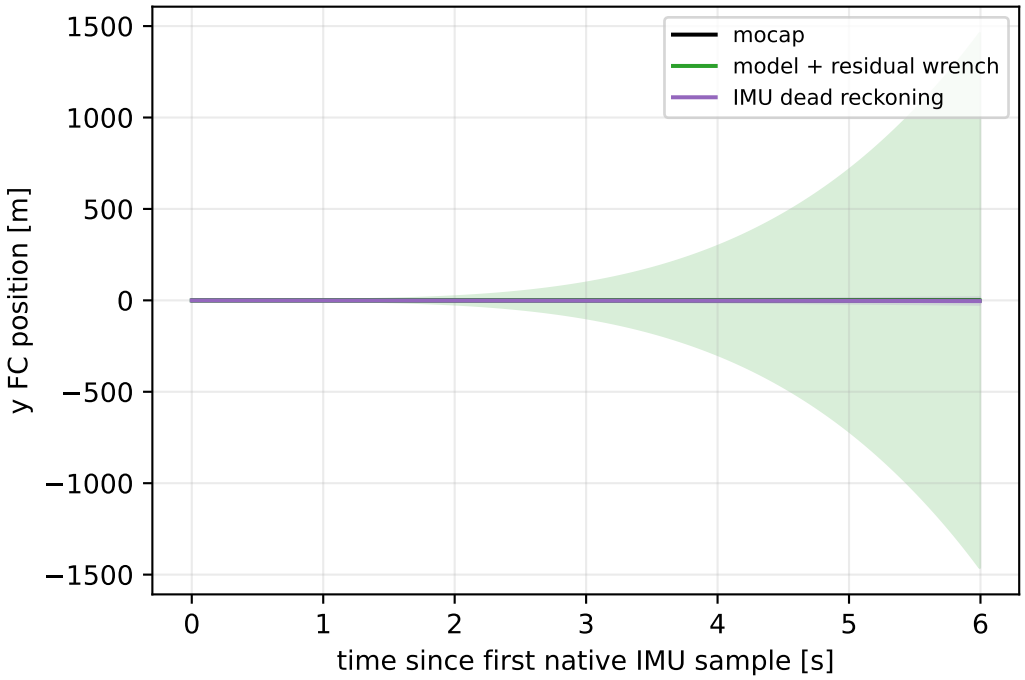


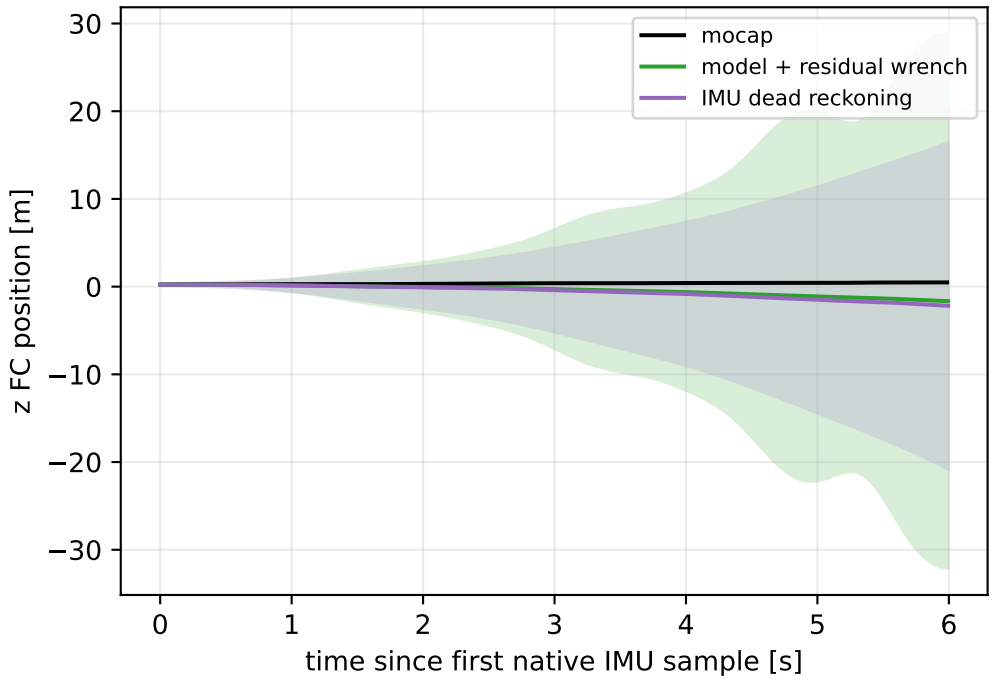


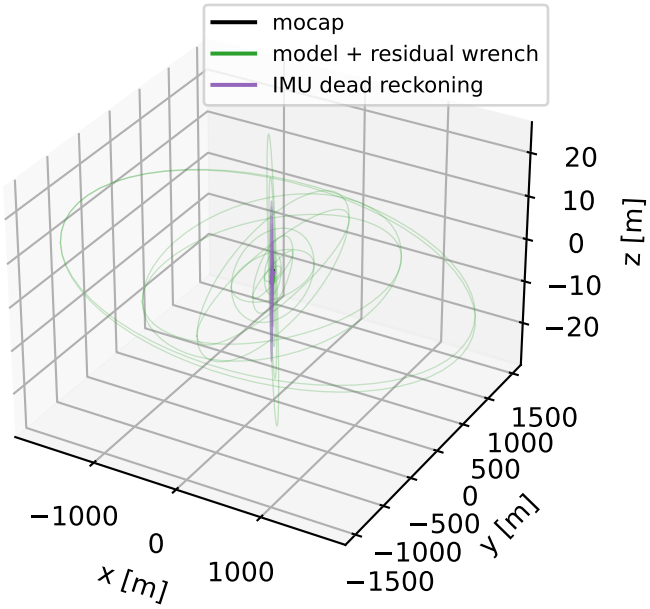
Residual-wrench model replay and IMU integration are not reset to later mocap.

failure1 — Position reconstruction with full propagated ±2σ spread









FC origin; local Gaussian linearization. Ellipsoids: deterministic 2σ factor-SVD wireframes, not samples.

failure2 — Native-IMU reconstruction: method and interpretation

Separate experiment: existing mocap identification and outputs are unchanged.

Identification uses native gyro / specific force with fitted thrust, gyro and accel first-order lags.

Orientation: gyro prediction + acceleration update recur; magnetic factors affect output only.

Only initial orientation / position and initial velocity use mocap. Later mocap is read-only evaluation.

Comparison uses original mocap timestamps inside IMU support; reconstructed attitudes are Slerped.

Native timestamps and firmware first-or-changed magnetic freshness proxy are retained.

Static sensor covariance = trace(unbiased static covariance)/3 times identity; no bias subtraction.

Offline local polynomial derivatives and their overlap covariance are included in the lag fit.

Uncertainty: Bingham-derived right-angle covariance and deterministic local Gaussian propagation.

Shared acceleration/attitude information uses declared conservative covariance bounds.

These algebraic bounds do not guarantee coverage of systematic acceleration or magnetic bias.

Magnetic output can be worse on these data; no mocap-based gating, tuning or recursive feedback.

Model reconstruction adds the native-IMU residual wrench time series to the physical dynamics.

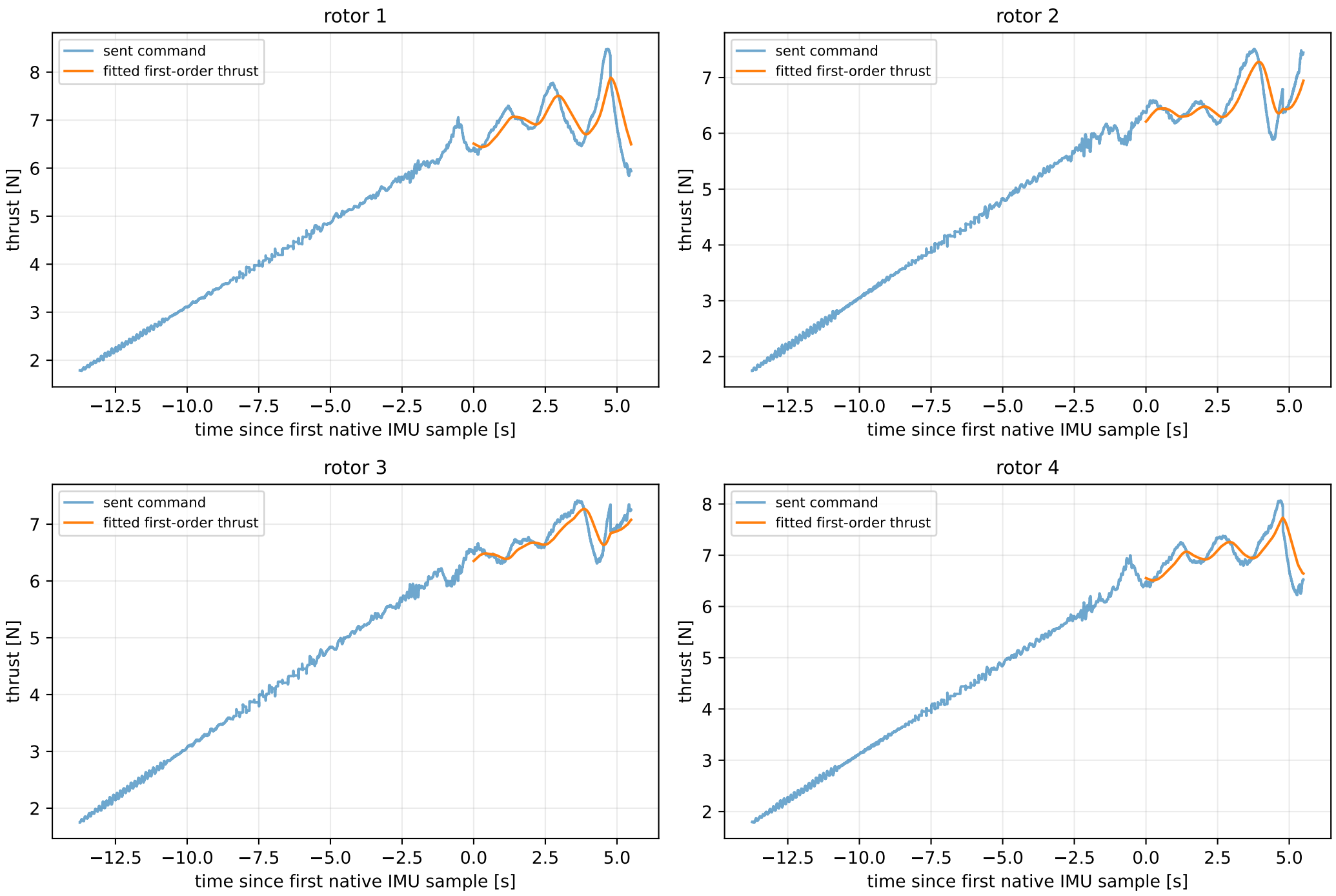
No Monte Carlo, artificial display variance, eigenvalue clipping or uncertainty-based sample selection.

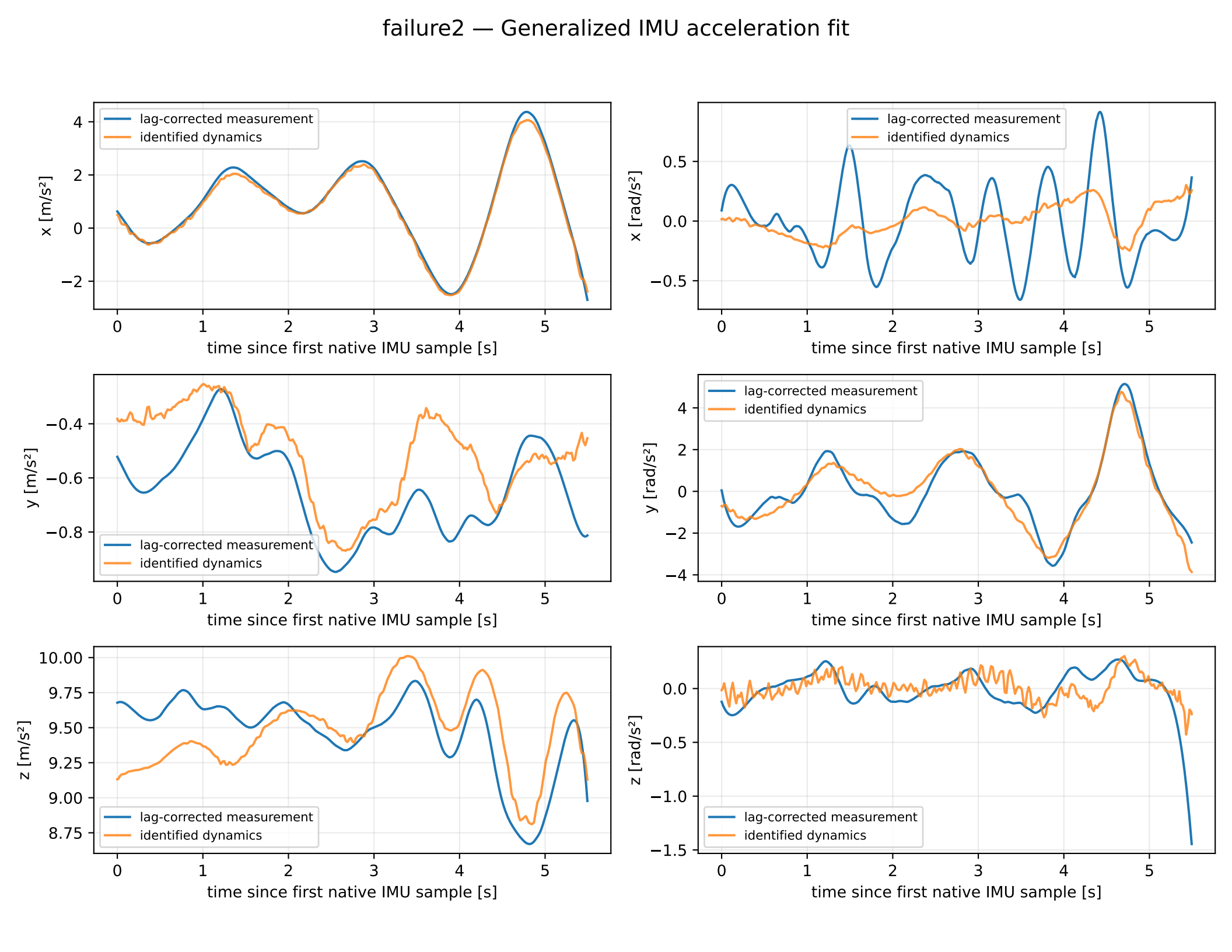
Mass / inertia / thrust scale use a common-scale gauge; chip lever arm is fixed, not calibrated.

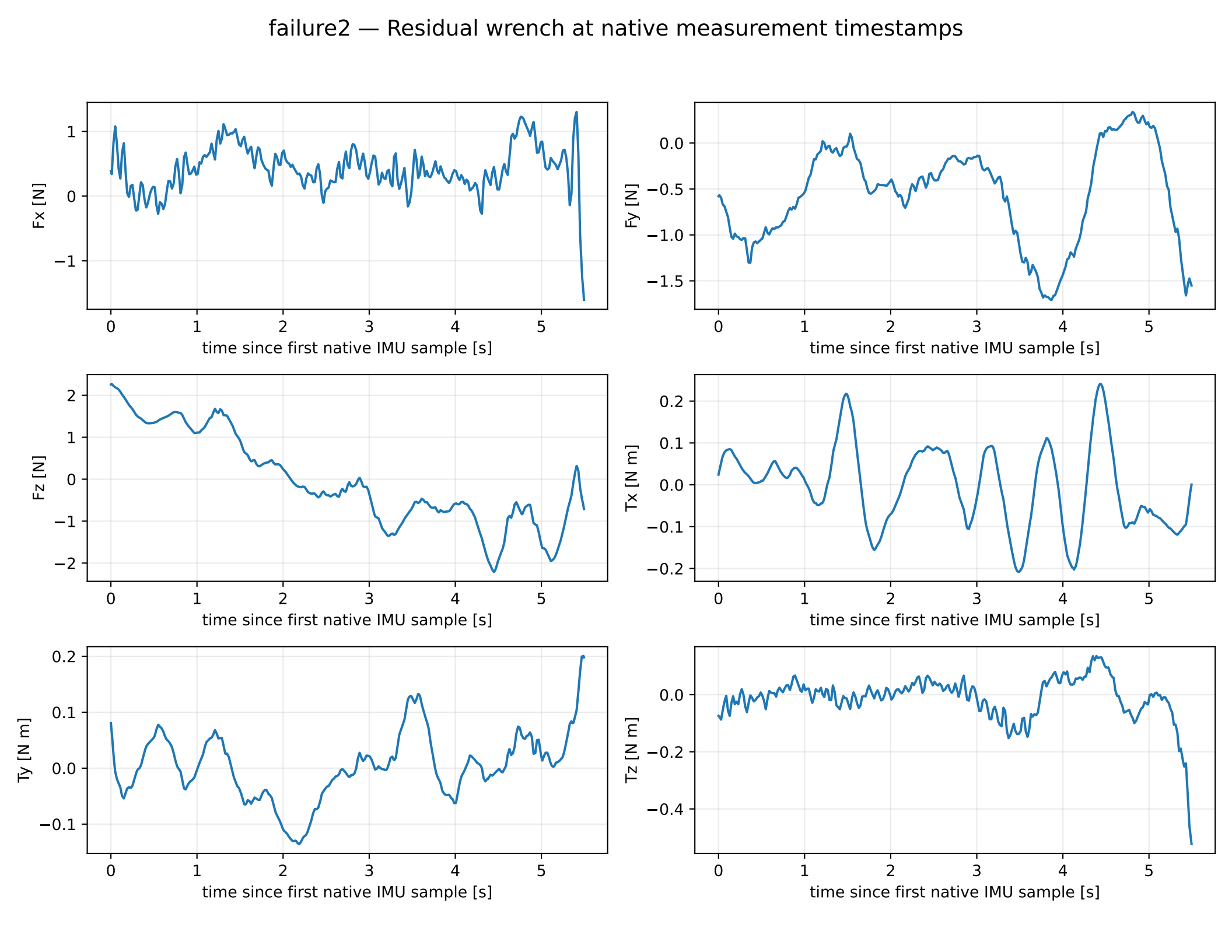
All 17 actual SI estimates and the complete joint covariance are appended at the end.

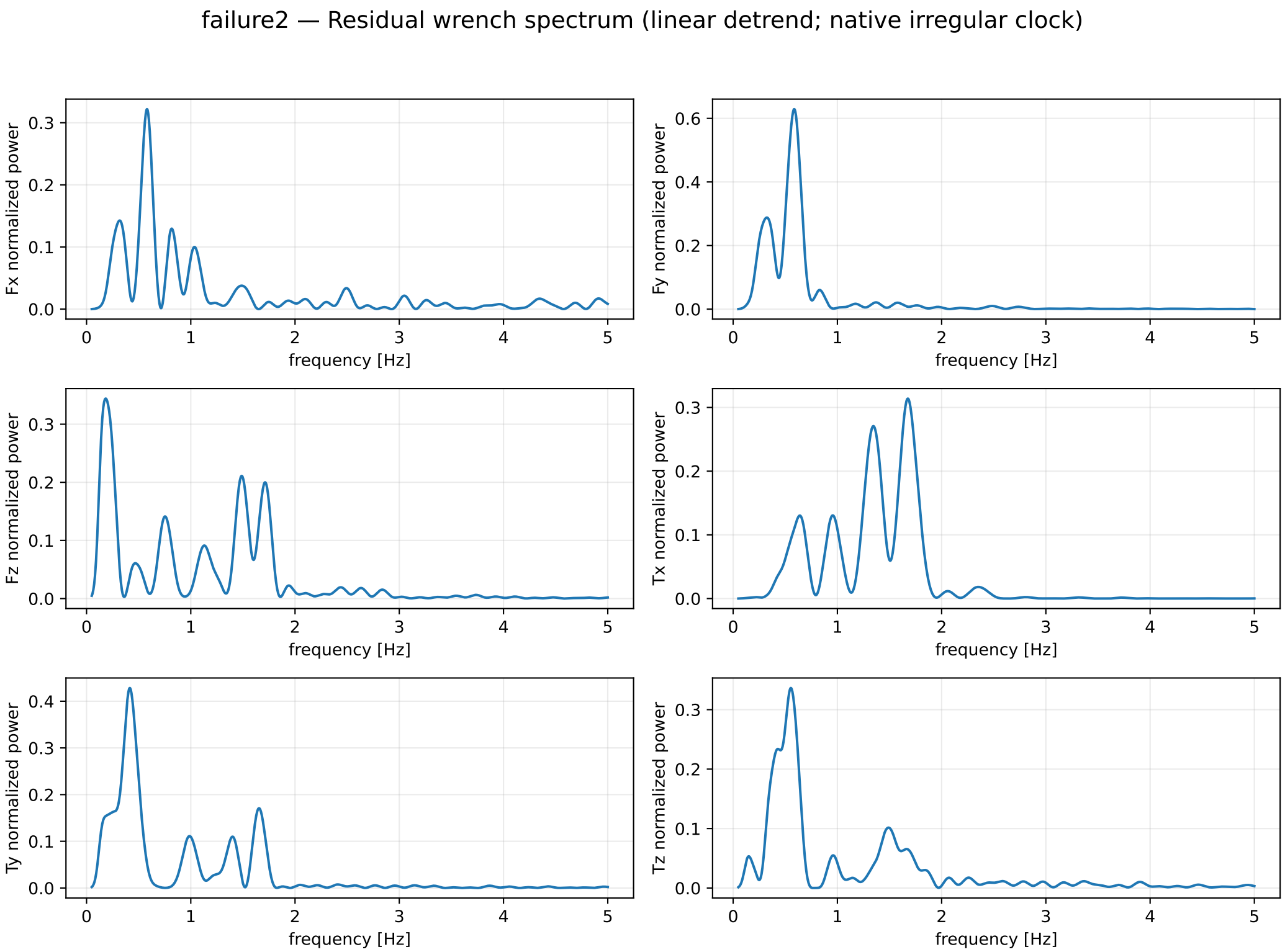
Native IMU support: 25.504054 to 30.998216 bag seconds; N = 1100.

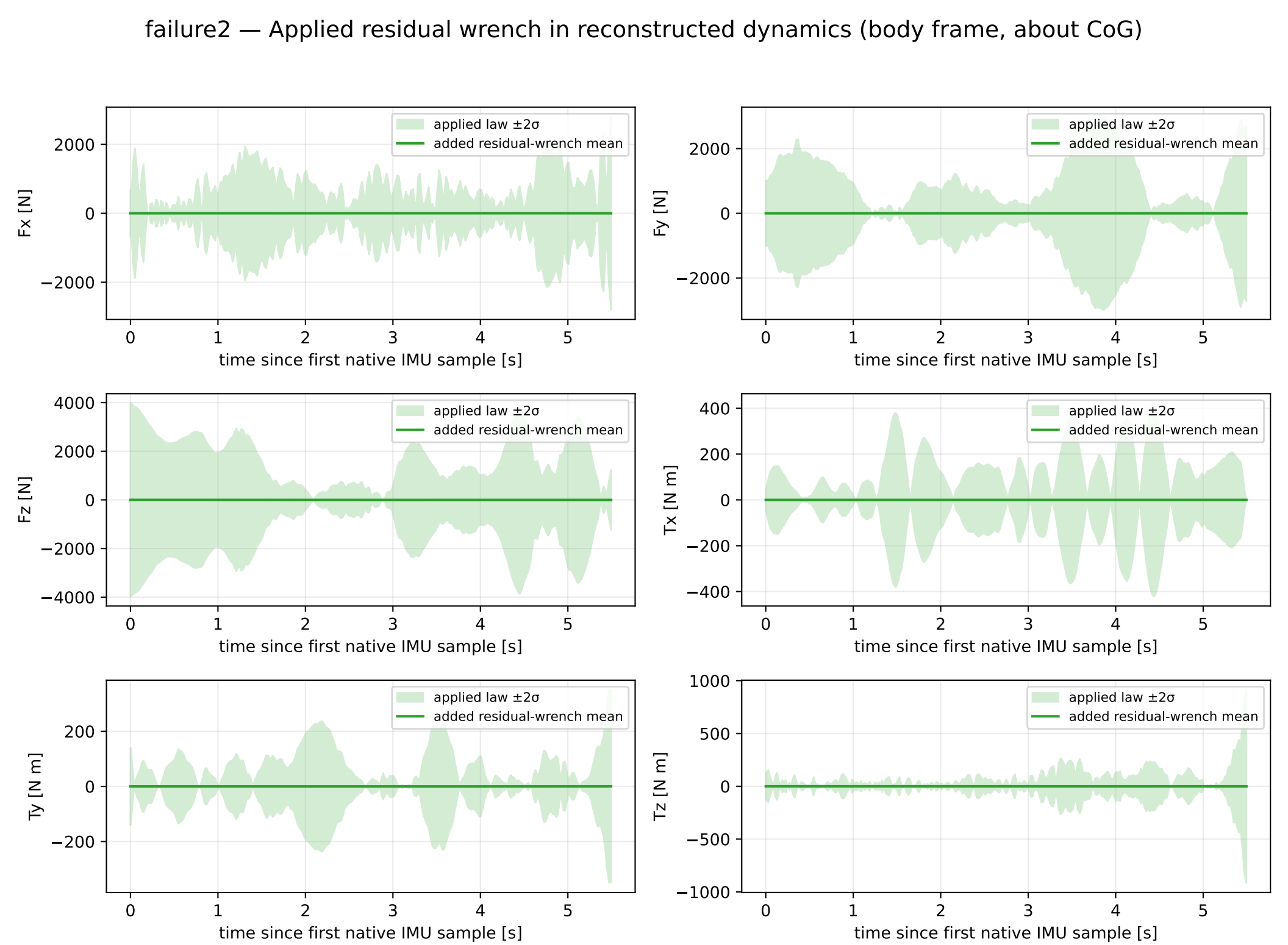
Individual PDF appendices and full-precision covariance CSV contain the same distribution.

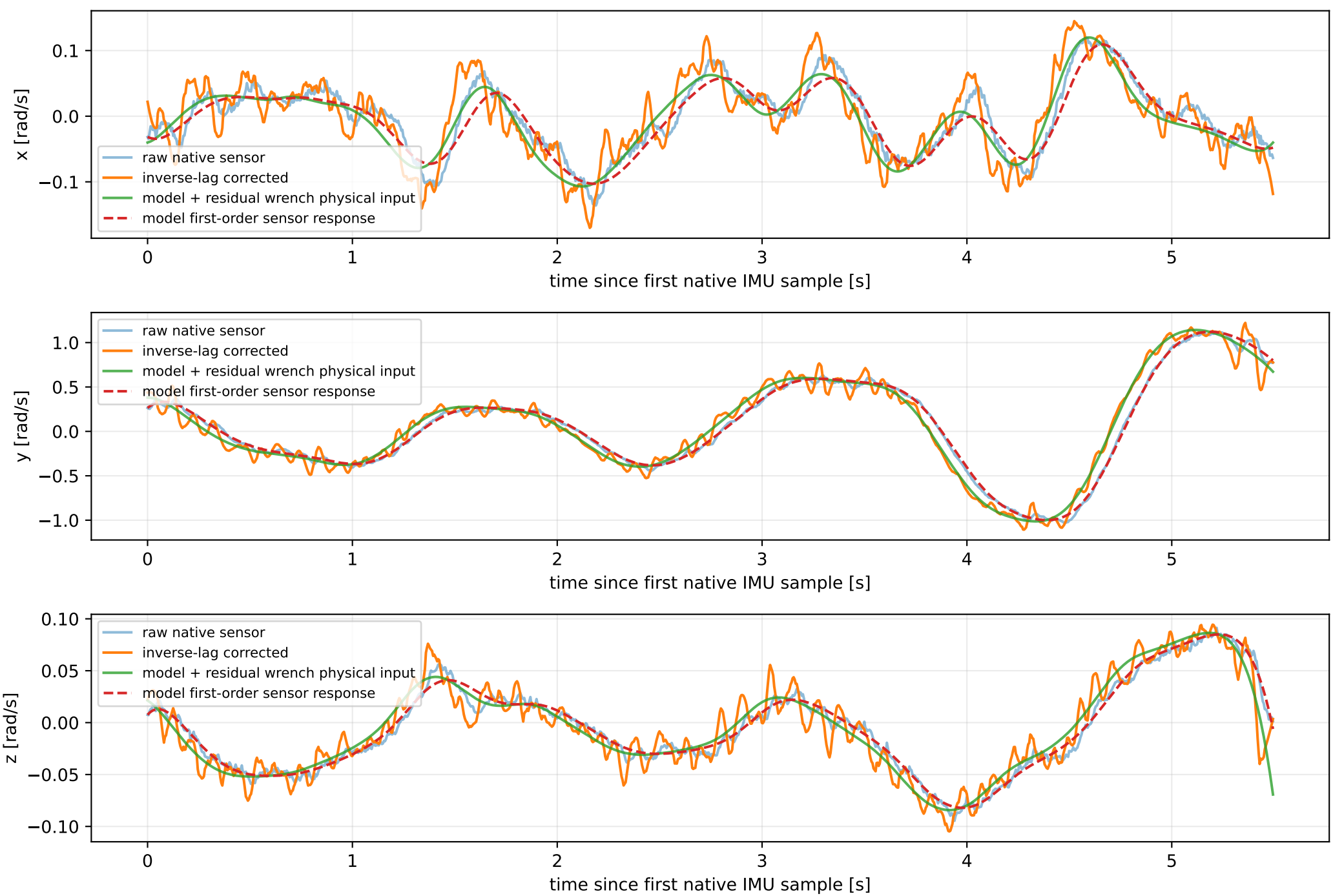

failure2 — Rotor commands and fitted thrust dynamics
rotor 1
rotor 2
rotor 3
rotor 4
sent command
fitted first-order thrust
thrust [N]
time since first native IMU sample [s]
−12.5
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0

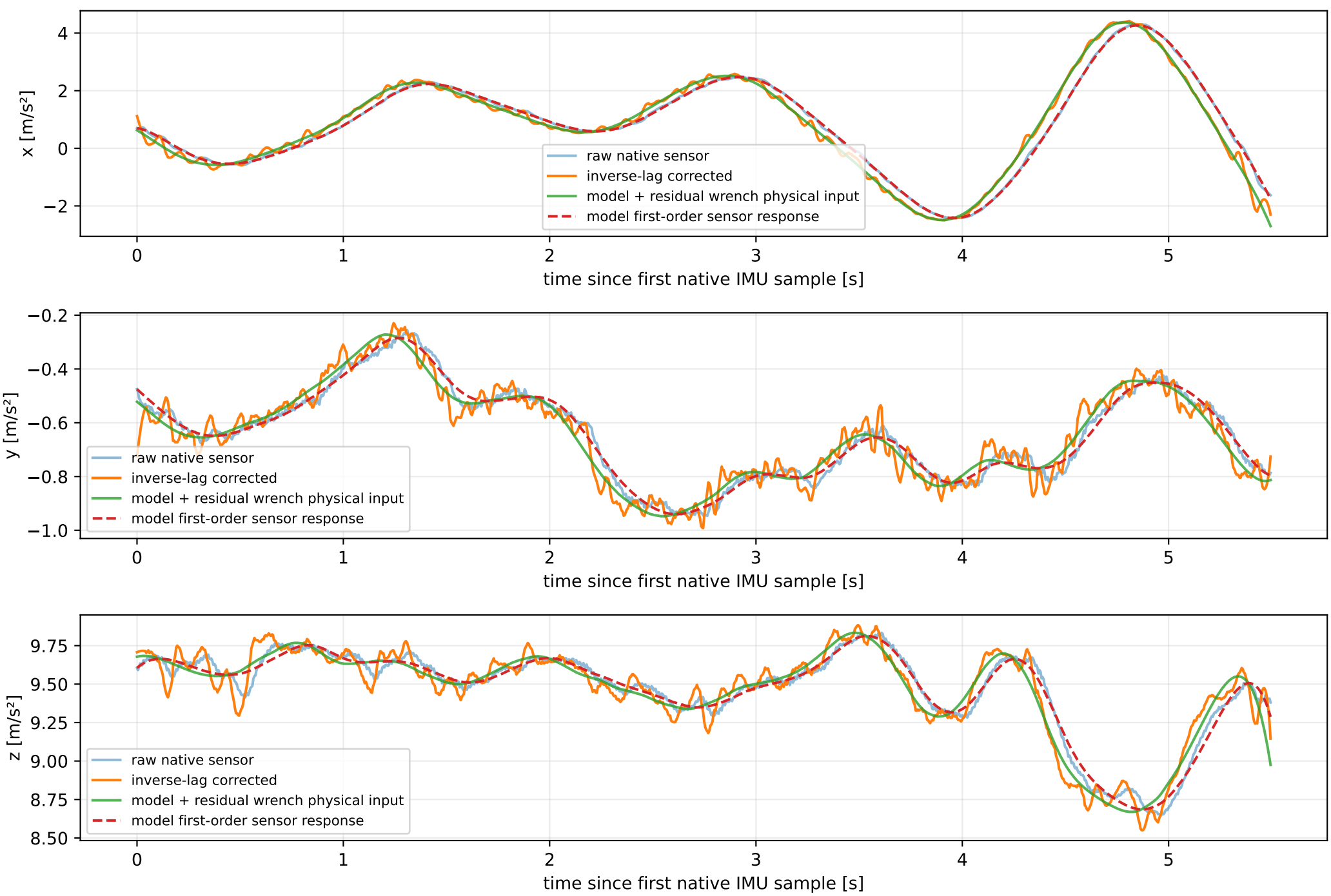

failure2 — Generalized IMU acceleration fit
lag-corrected measurement
identified dynamics
x [m/s²]
x [rad/s²]
y [m/s²]
y [rad/s²]
z [m/s²]
z [rad/s²]
time since first native IMU sample [s]

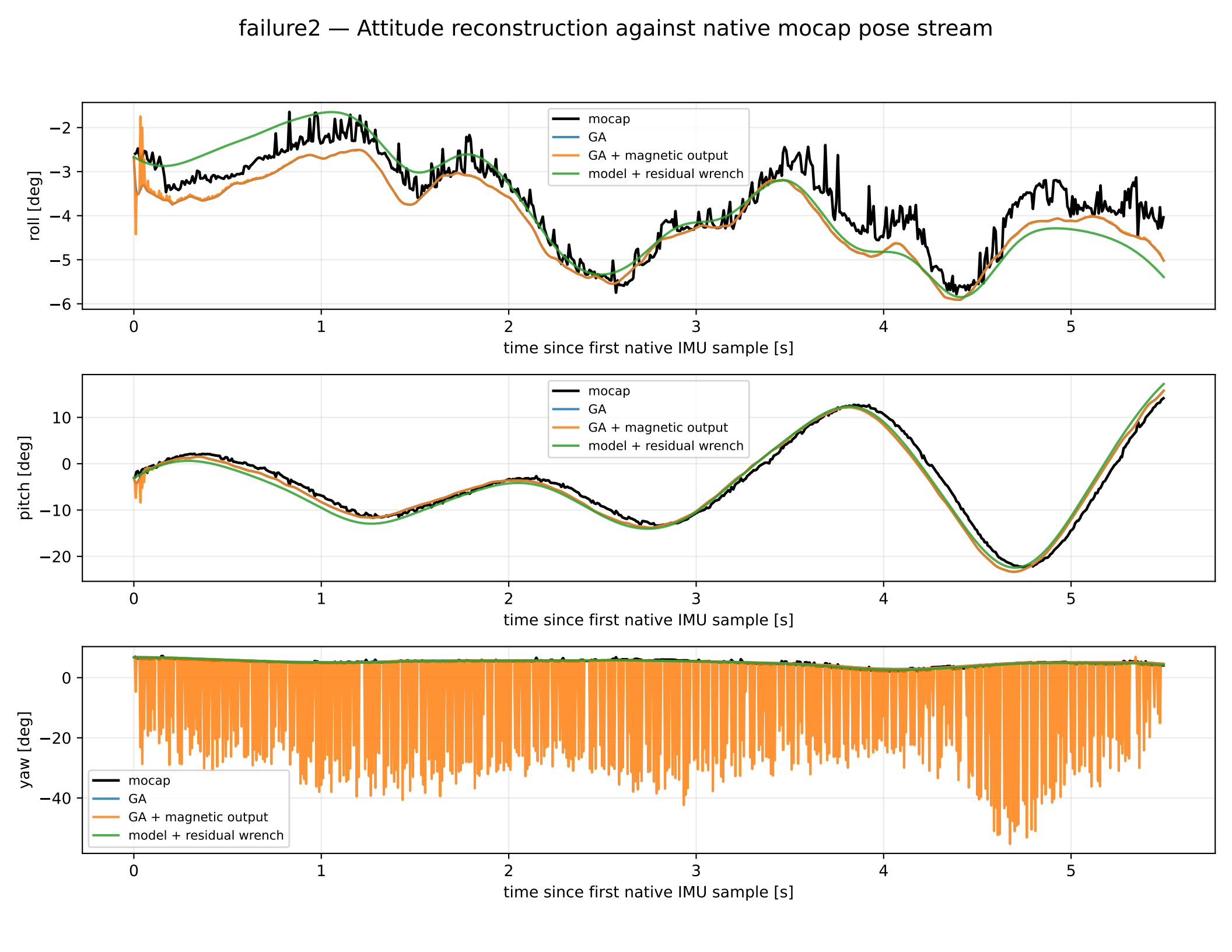

failure2 — Residual wrench at native measurement timestamps
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
time since first native IMU sample [s]

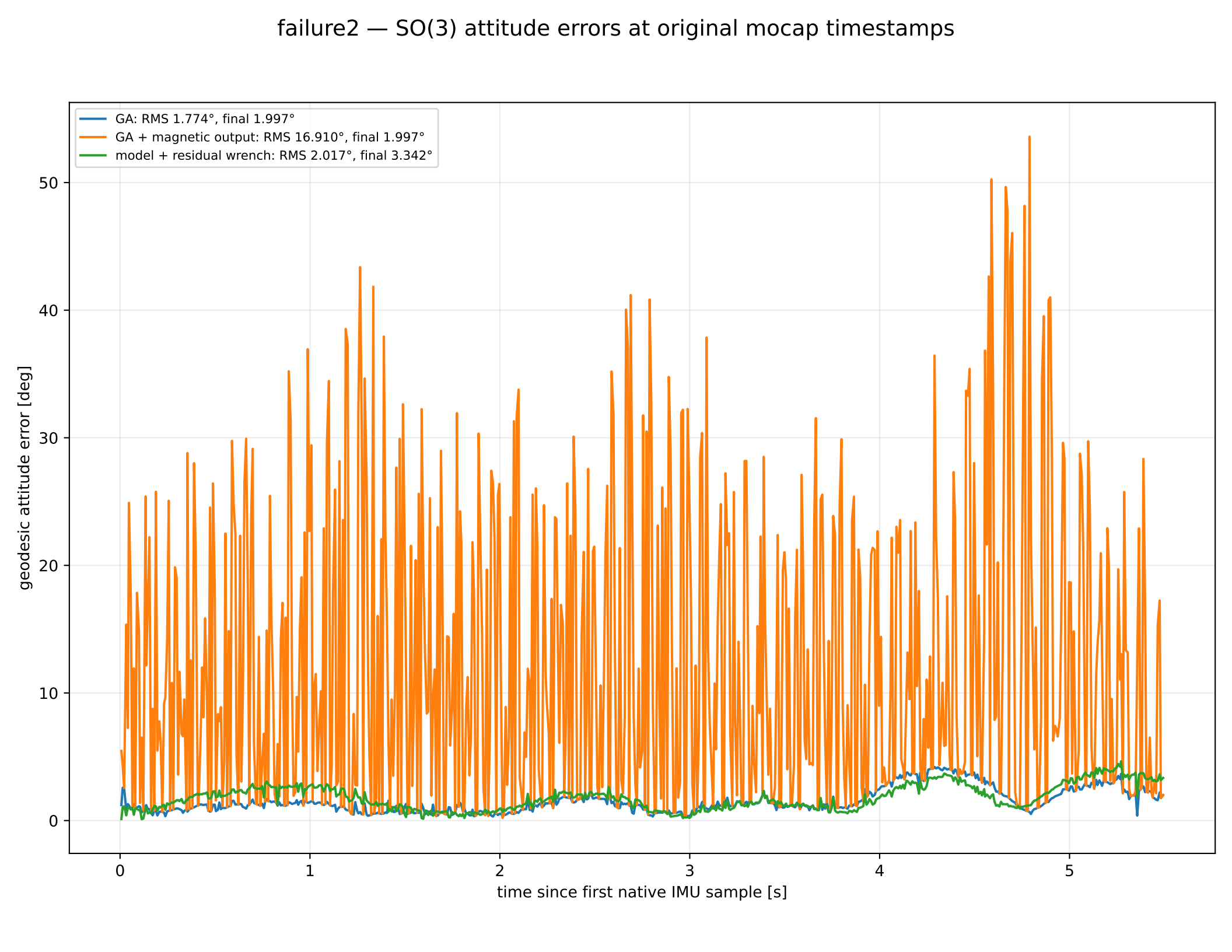

failure2 — Residual wrench spectrum (linear detrend; native irregular clock)
Fx normalized power
Fy normalized power
Fz normalized power
Tx normalized power
Ty normalized power
Tz normalized power
frequency [Hz]

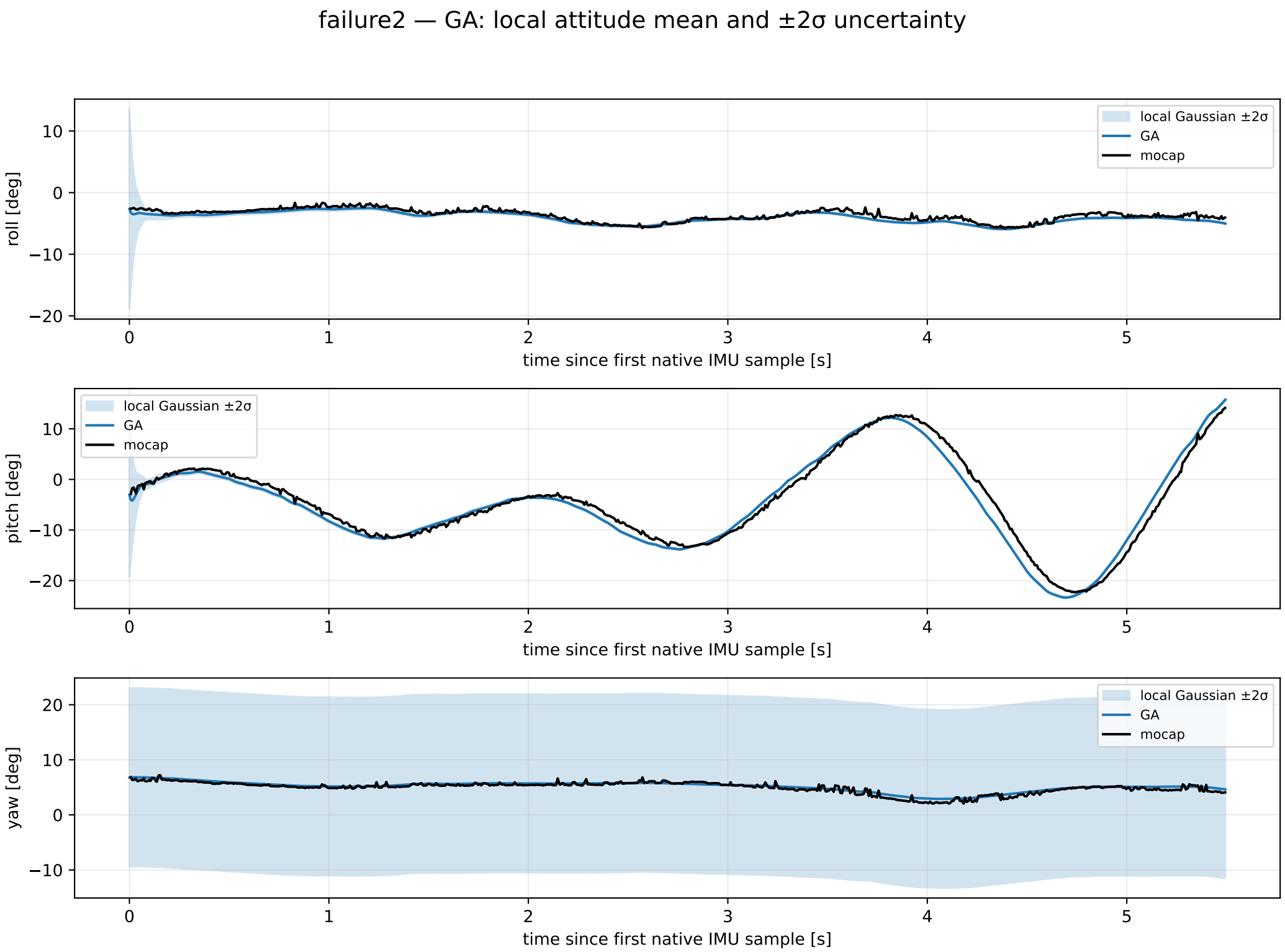

failure2 — Applied residual wrench in reconstructed dynamics (body frame, about CoG)
applied law ±2σ
added residual-wrench mean
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
time since first native IMU sample [s]
The native-IMU residual wrench is added to the dynamics; this is not a model-only free replay.

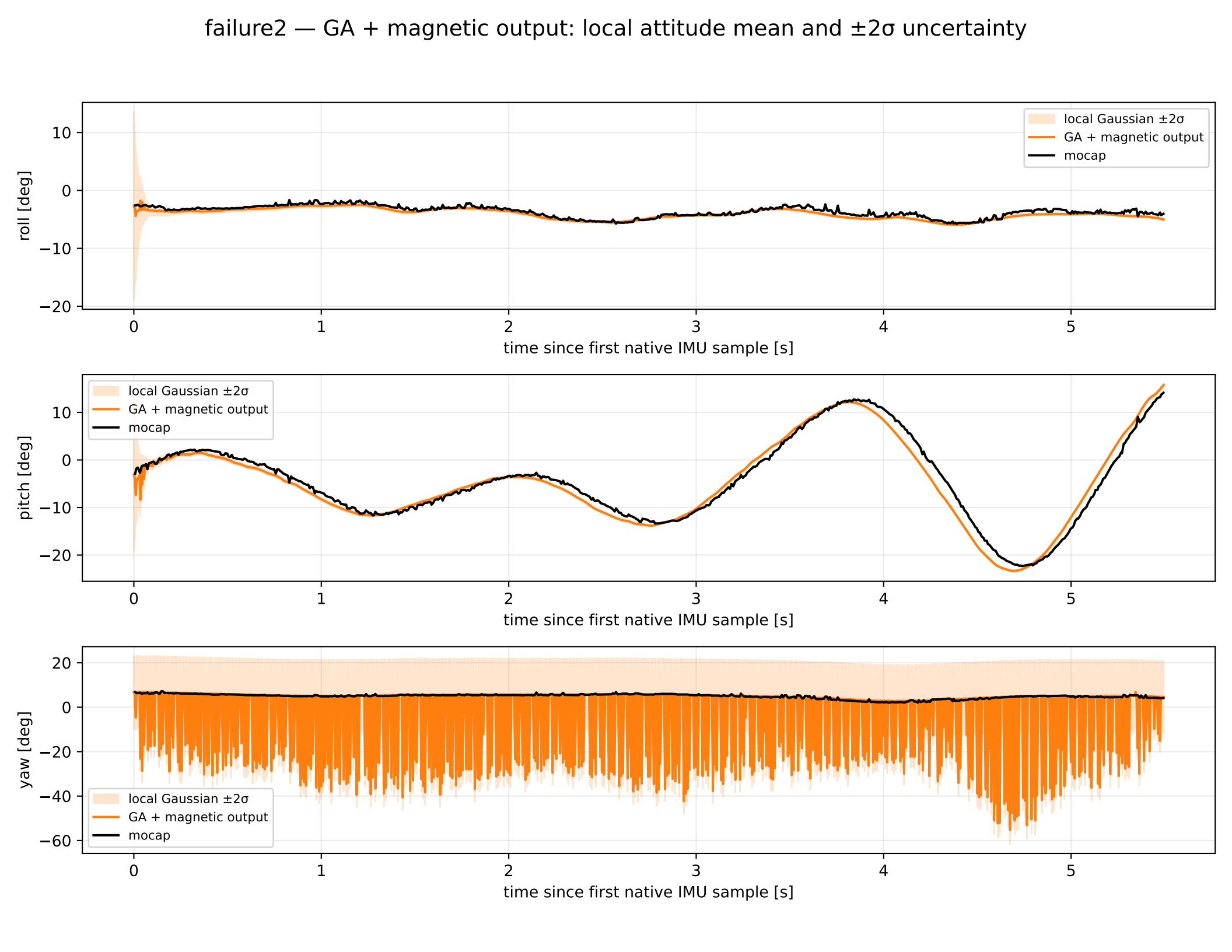

failure2 — gyro: raw, inverse-lag correction and model measurement response
x [rad/s]
y [rad/s]
z [rad/s]
raw native sensor
inverse-lag corrected
model + residual wrench physical input
model first-order sensor response
time since first native IMU sample [s]

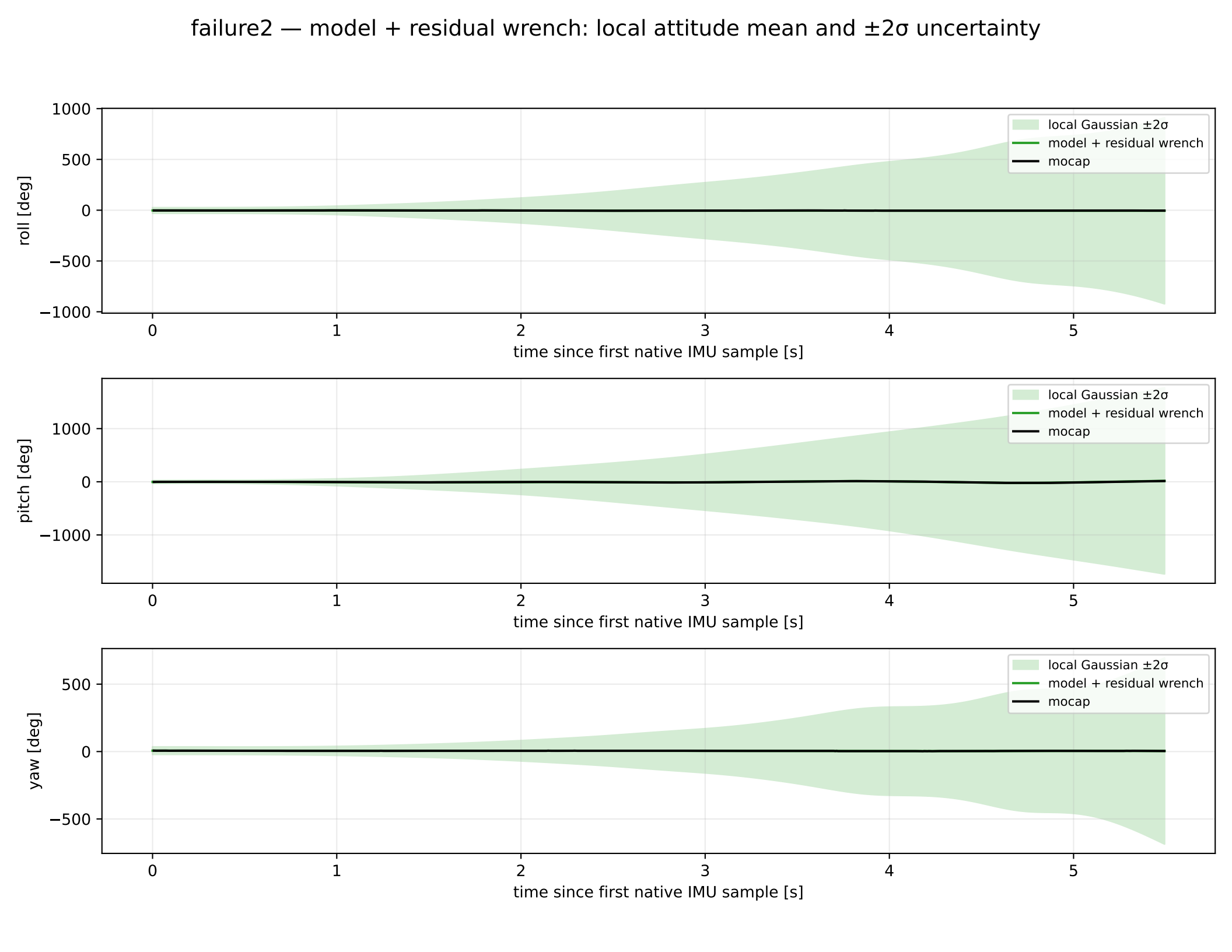

failure2 — accel: raw, inverse-lag correction and model measurement response
x [m/s²]
y [m/s²]
z [m/s²]
time since first native IMU sample [s]
raw native sensor
inverse-lag corrected
model + residual wrench physical input
model first-order sensor response


failure2 — Attitude reconstruction against native mocap pose stream
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
mocap
GA
GA + magnetic output
model + residual wrench

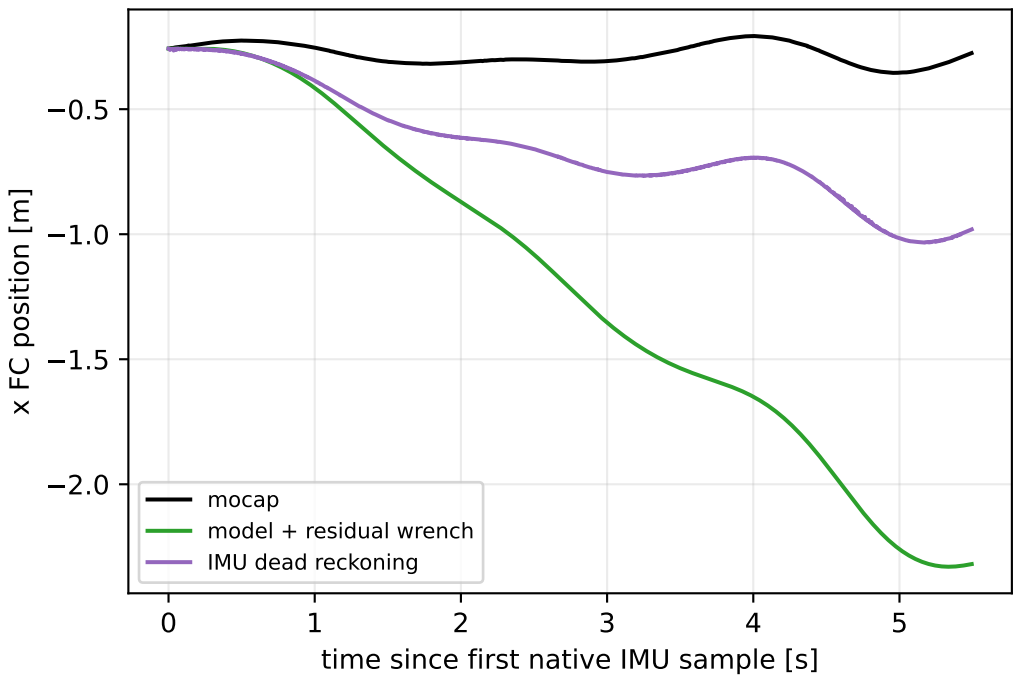

failure2 — SO(3) attitude errors at original mocap timestamps
GA: RMS 1.774°, final 1.997°
GA + magnetic output: RMS 16.910°, final 1.997°
model + residual wrench: RMS 2.017°, final 3.342°
geodesic attitude error [deg]
time since first native IMU sample [s]
0
10
20
30
40
50
0
1
2
3
4
5

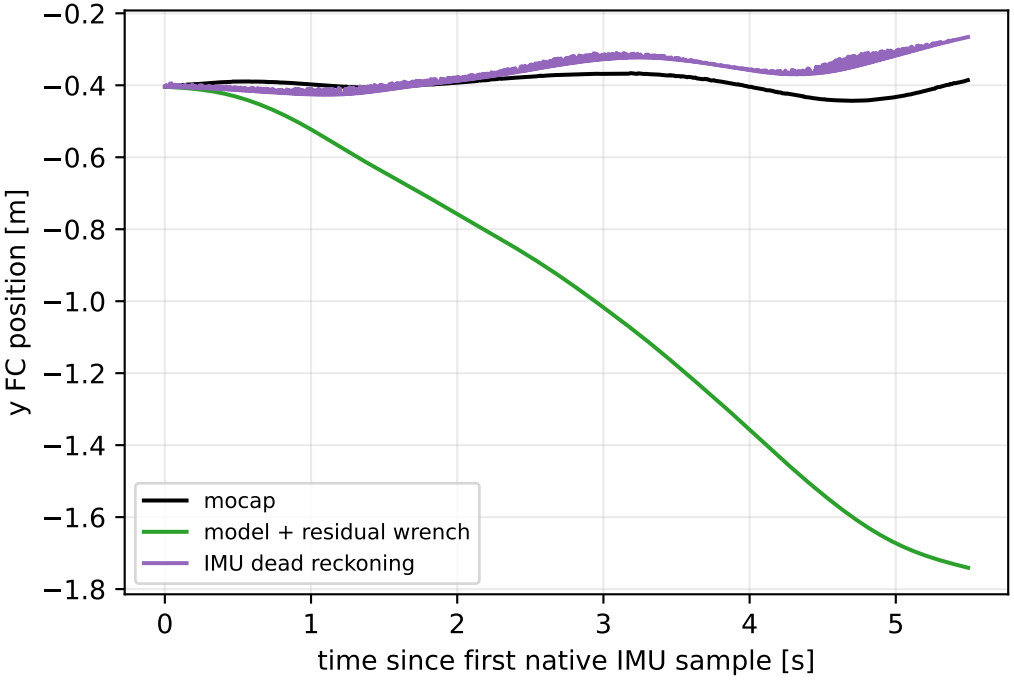

failure2 — GA: local attitude mean and ±2σ uncertainty
local Gaussian ±2σ
GA
mocap
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]

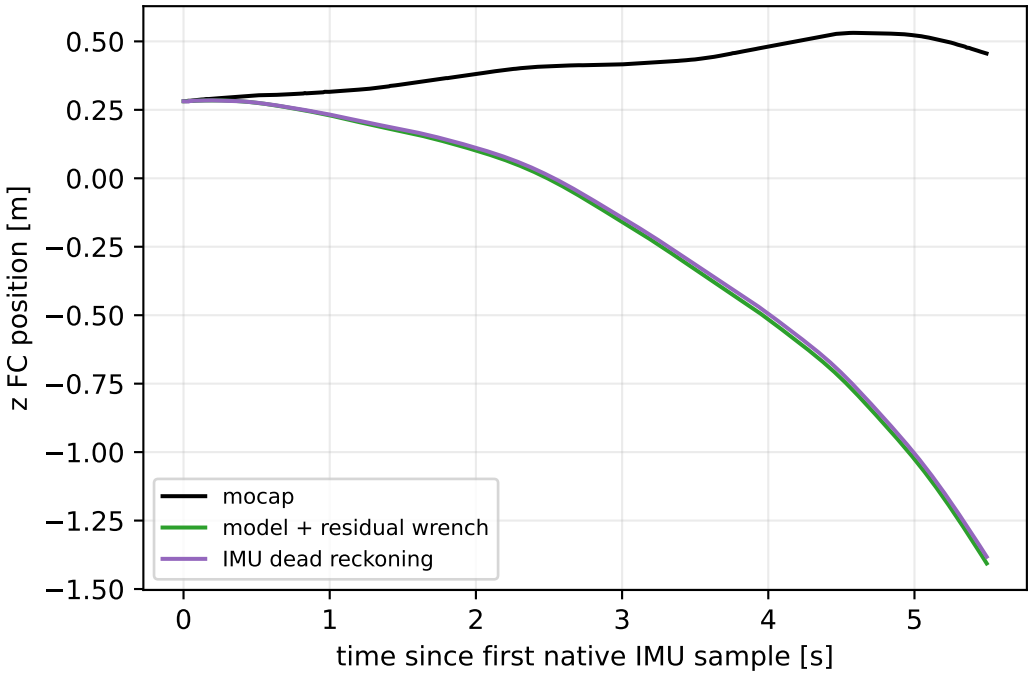

failure2 — GA + magnetic output: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA + magnetic output
mocap

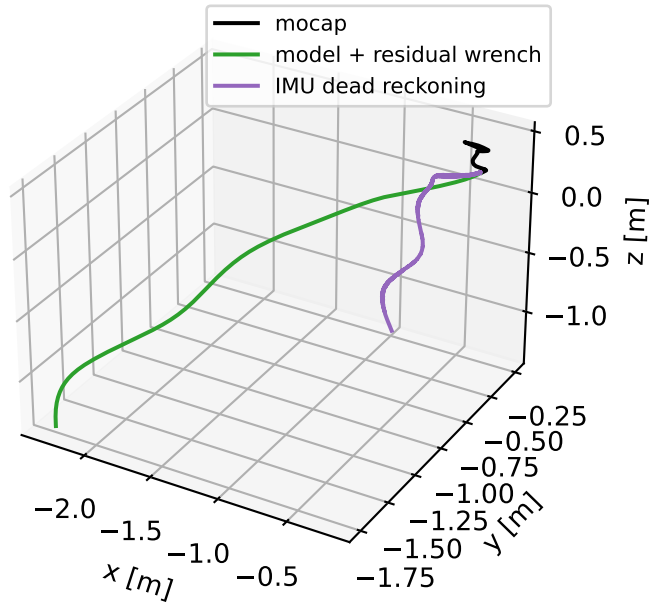

failure2 — model + residual wrench: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
model + residual wrench
mocap

failure2 — Position reconstruction (mean trajectories)



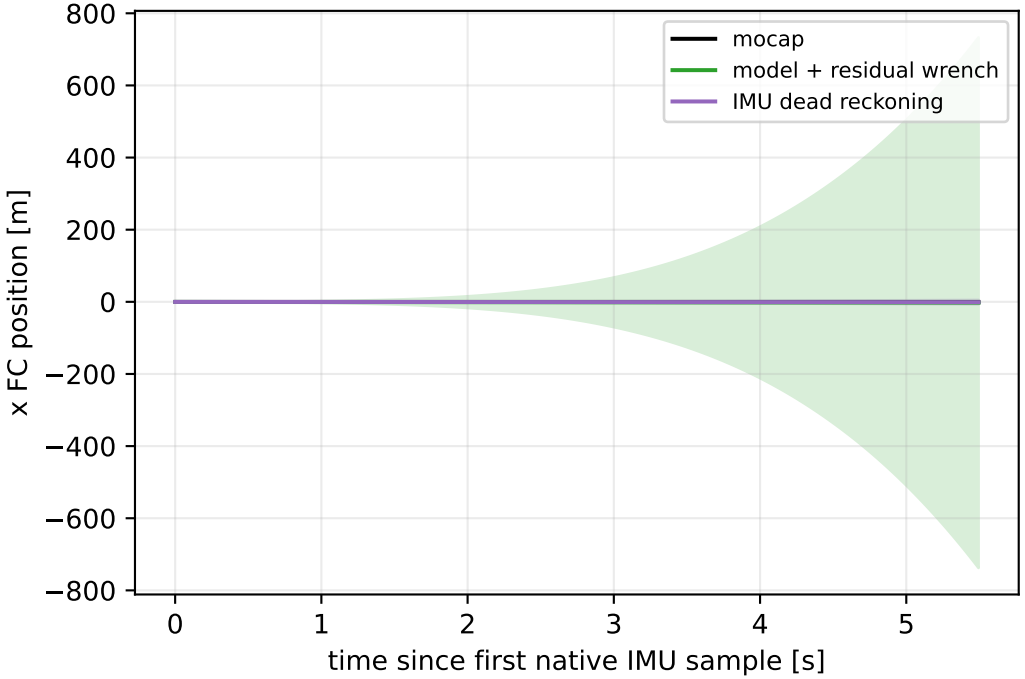


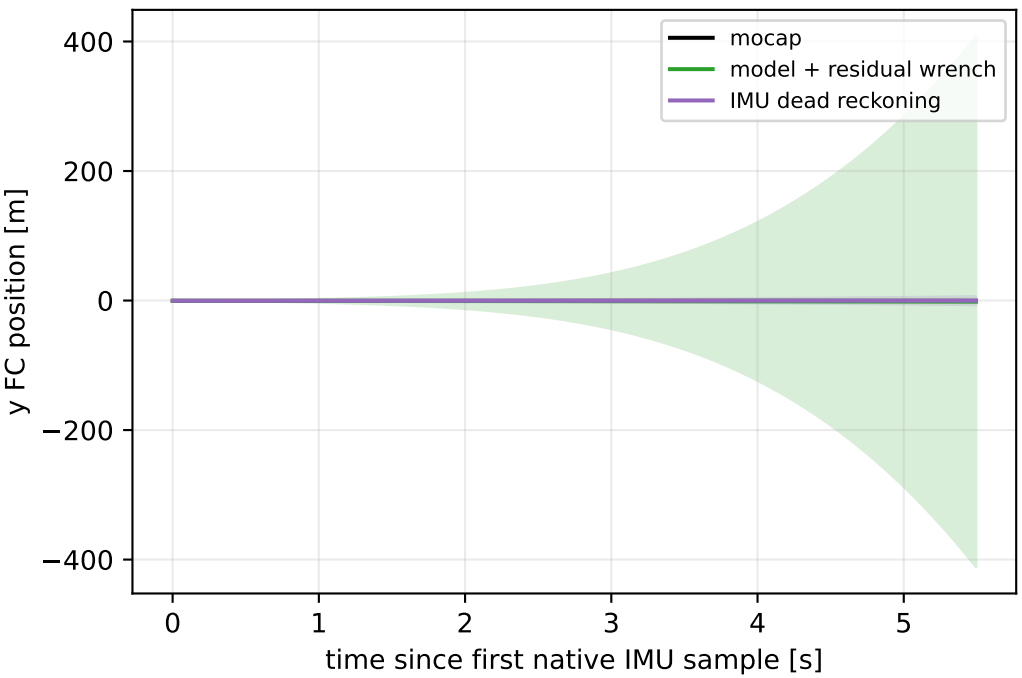


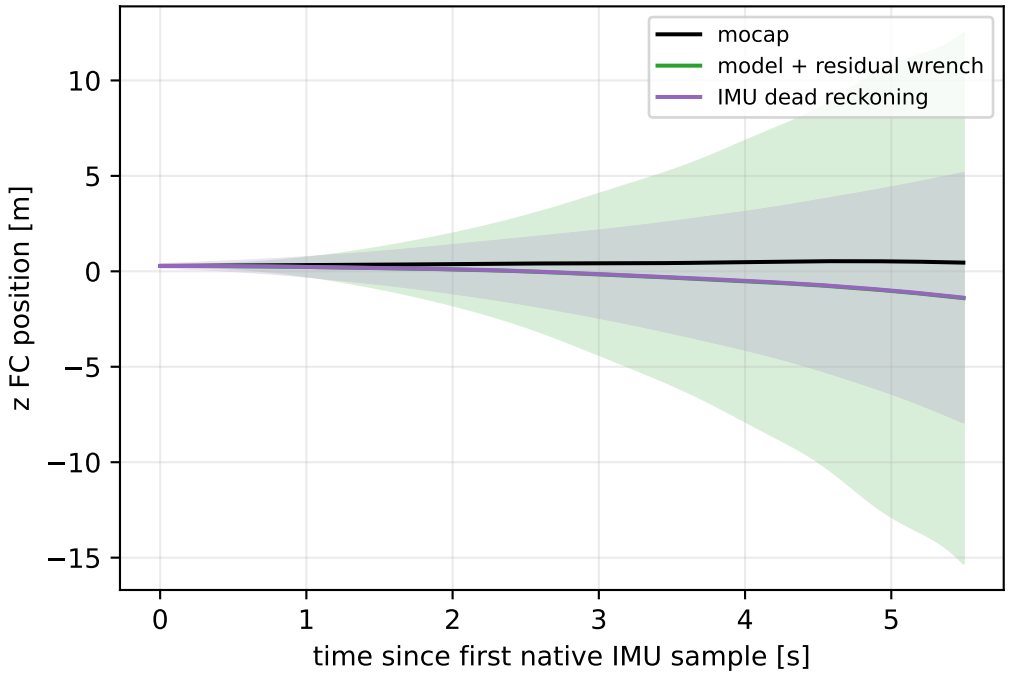


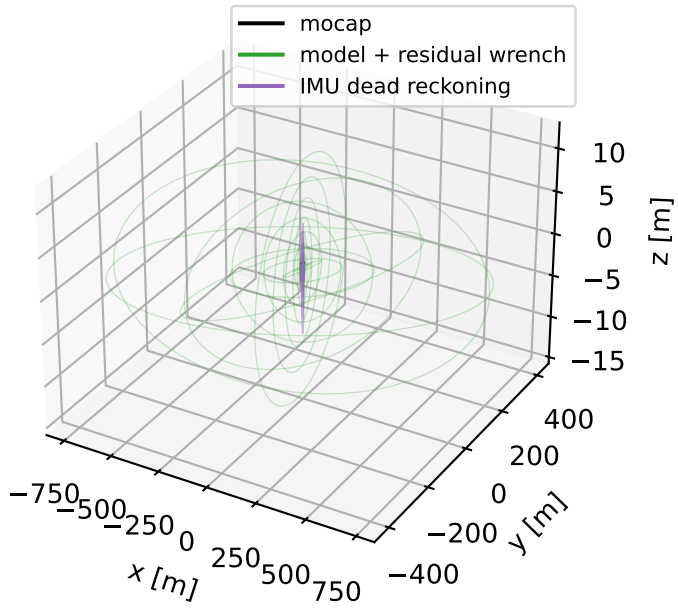
Residual-wrench model replay and IMU integration are not reset to later mocap.

failure2 — Position reconstruction with full propagated ±2σ spread









FC origin; local Gaussian linearization. Ellipsoids: deterministic 2σ factor-SVD wireframes, not samples.

success — Native-IMU reconstruction: method and interpretation

Separate experiment: existing mocap identification and outputs are unchanged.

Identification uses native gyro / specific force with fitted thrust, gyro and accel first-order lags.

Orientation: gyro prediction + acceleration update recur; magnetic factors affect output only.

Only initial orientation / position and initial velocity use mocap. Later mocap is read-only evaluation.

Comparison uses original mocap timestamps inside IMU support; reconstructed attitudes are Slerped.

Native timestamps and firmware first-or-changed magnetic freshness proxy are retained.

Static sensor covariance = trace(unbiased static covariance)/3 times identity; no bias subtraction.

Offline local polynomial derivatives and their overlap covariance are included in the lag fit.

Uncertainty: Bingham-derived right-angle covariance and deterministic local Gaussian propagation.

Shared acceleration/attitude information uses declared conservative covariance bounds.

These algebraic bounds do not guarantee coverage of systematic acceleration or magnetic bias.

Magnetic output can be worse on these data; no mocap-based gating, tuning or recursive feedback.

Model reconstruction adds the native-IMU residual wrench time series to the physical dynamics.

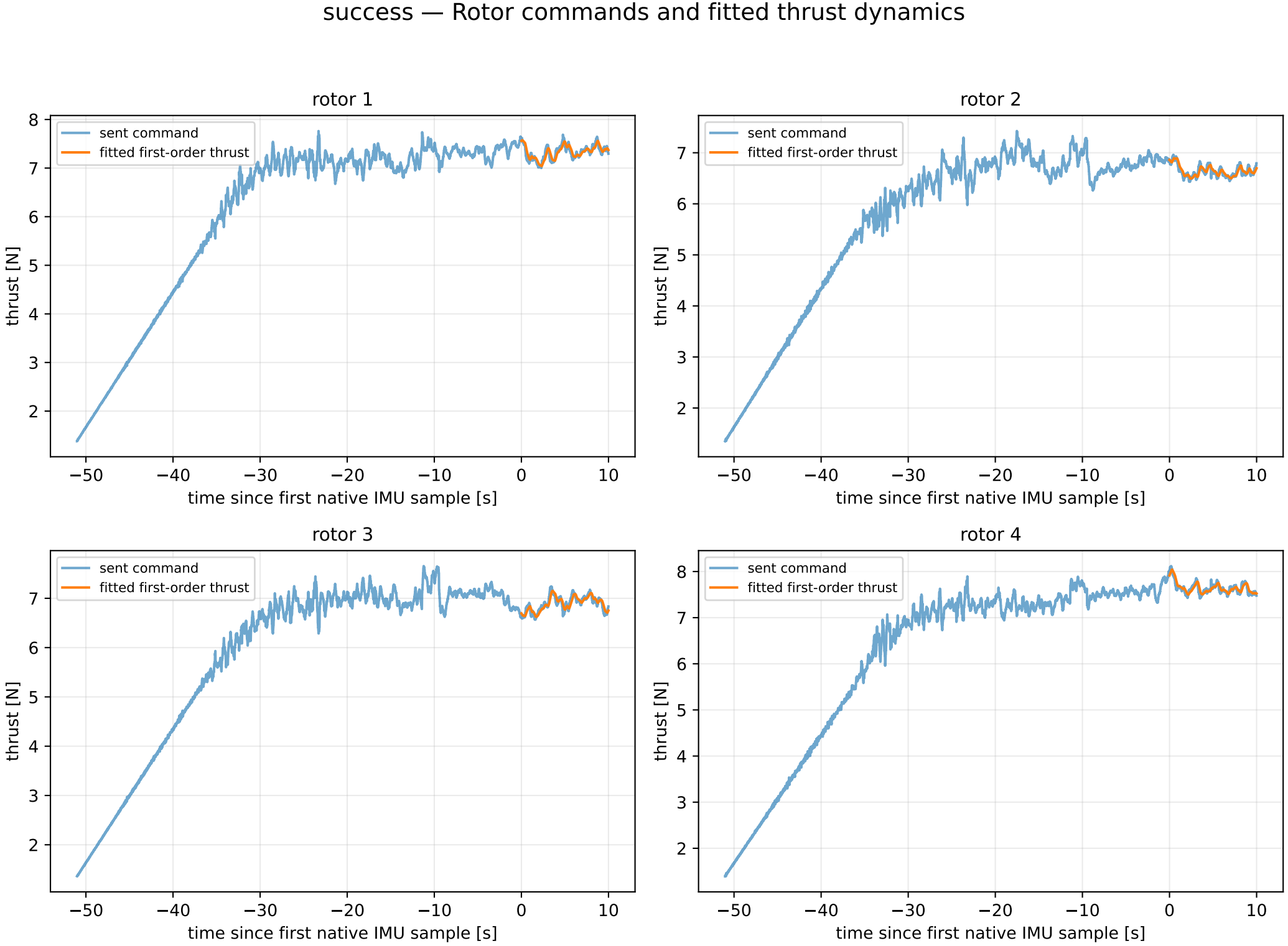
No Monte Carlo, artificial display variance, eigenvalue clipping or uncertainty-based sample selection.

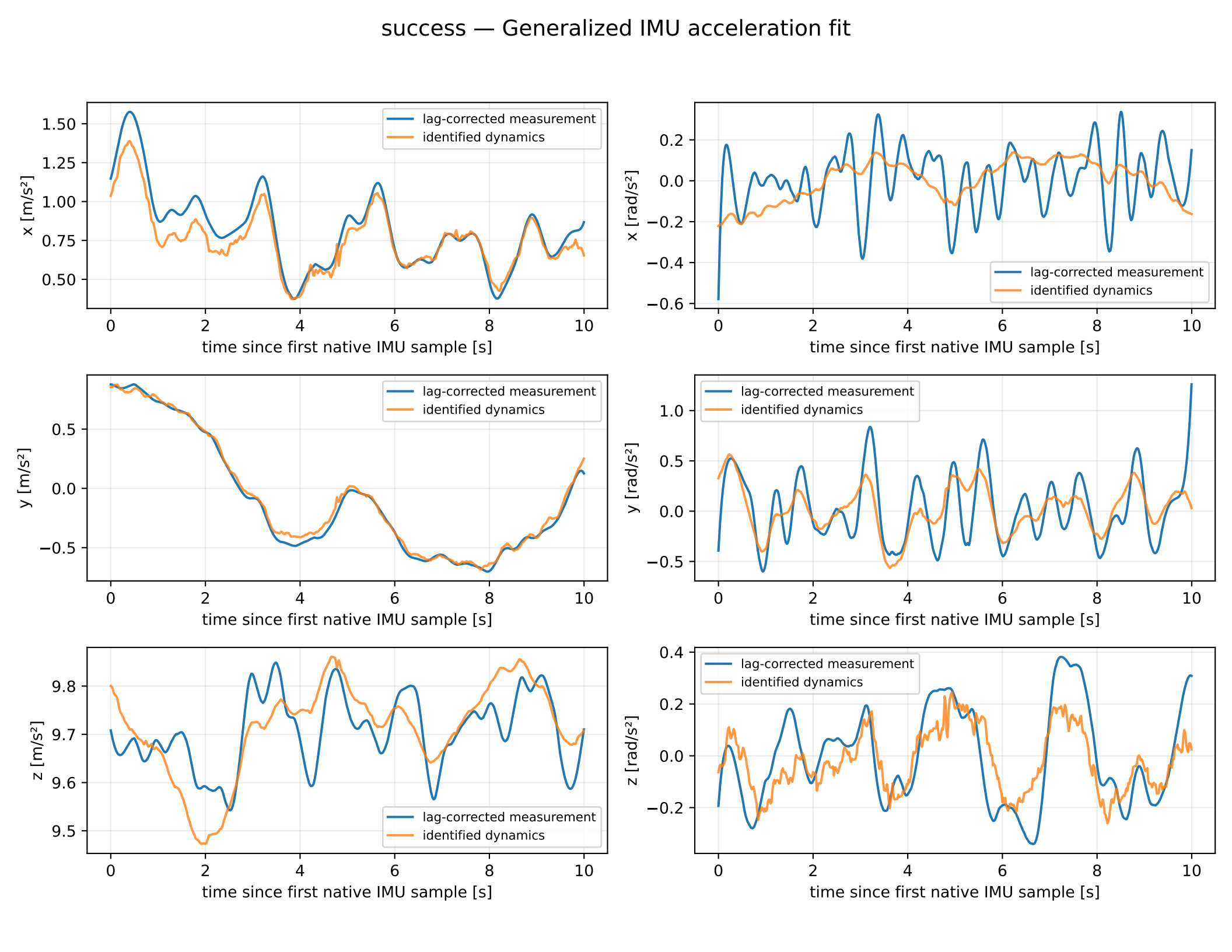
Mass / inertia / thrust scale use a common-scale gauge; chip lever arm is fixed, not calibrated.

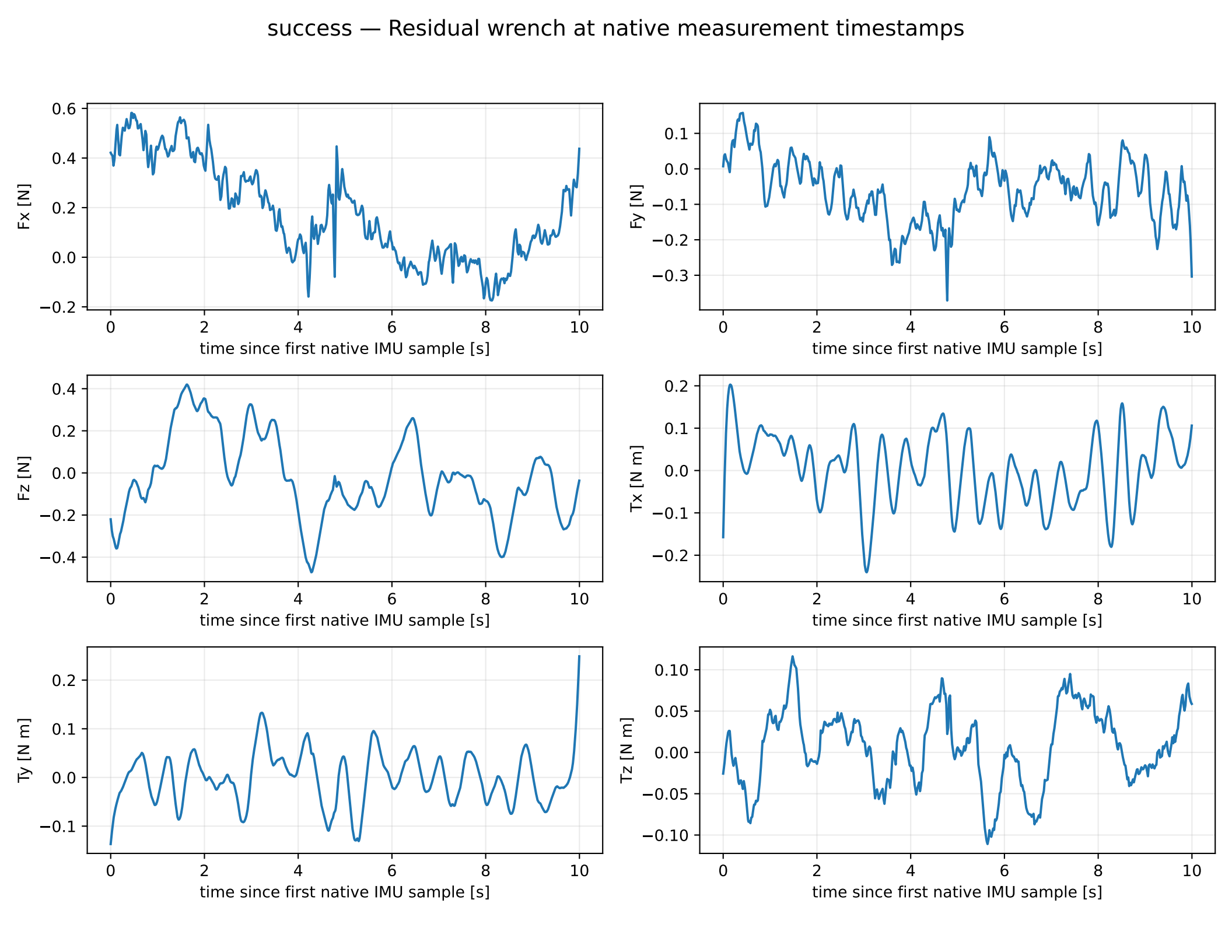
All 17 actual SI estimates and the complete joint covariance are appended at the end.

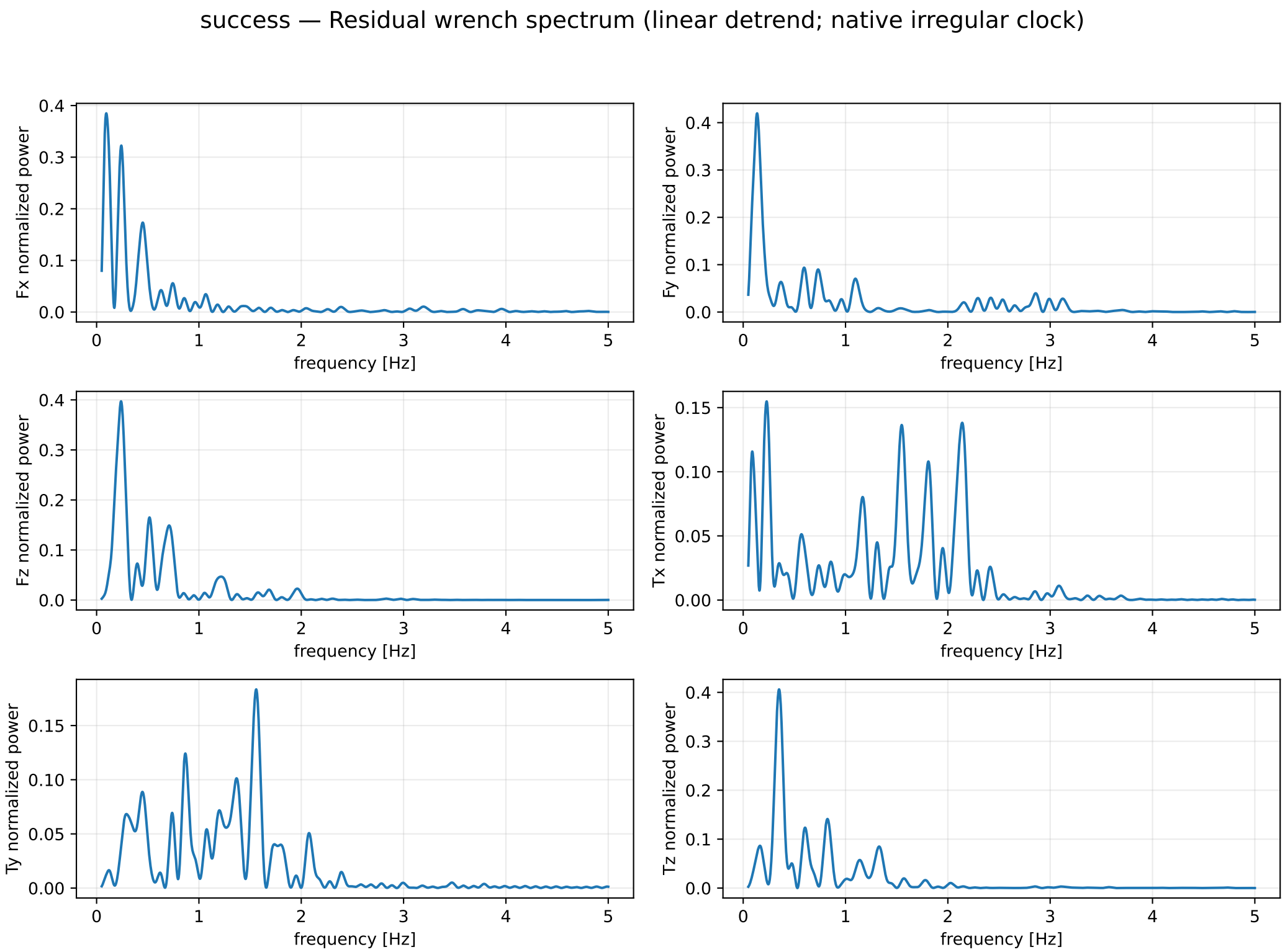
Native IMU support: 65.001922 to 74.998157 bag seconds; N = 2000.

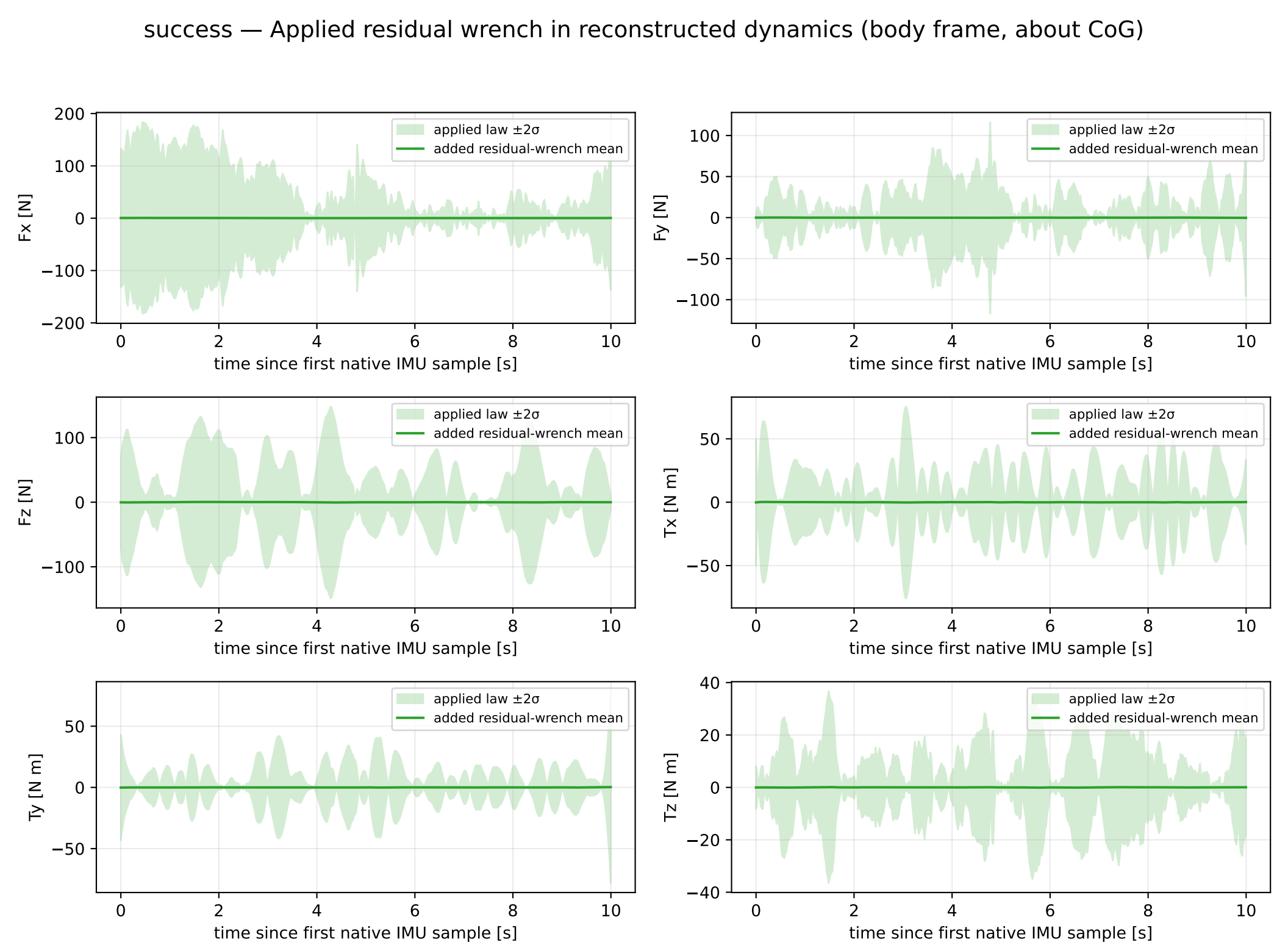
Individual PDF appendices and full-precision covariance CSV contain the same distribution.

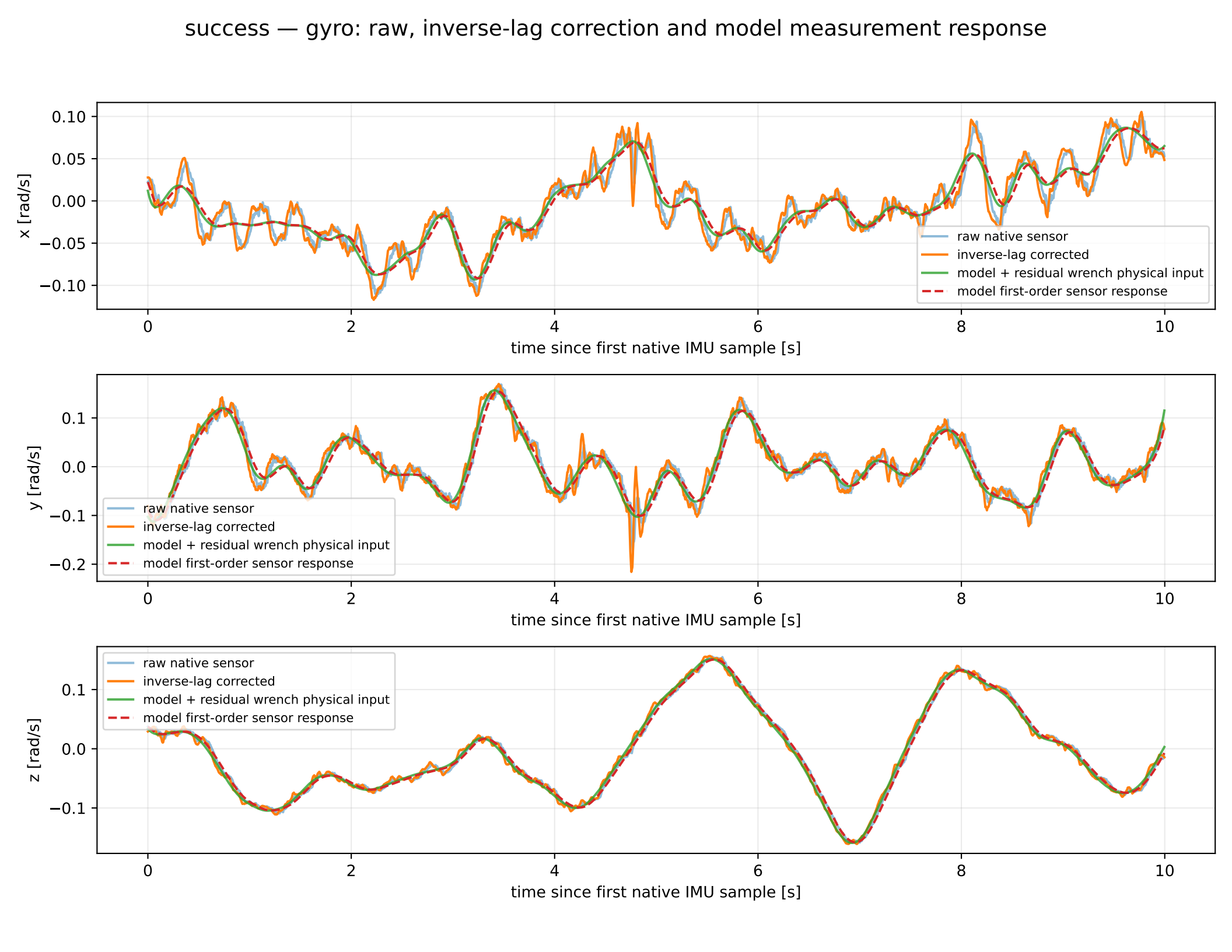

success — Rotor commands and fitted thrust dynamics
rotor 1
rotor 2
rotor 3
rotor 4
sent command
fitted first-order thrust
thrust [N]
time since first native IMU sample [s]

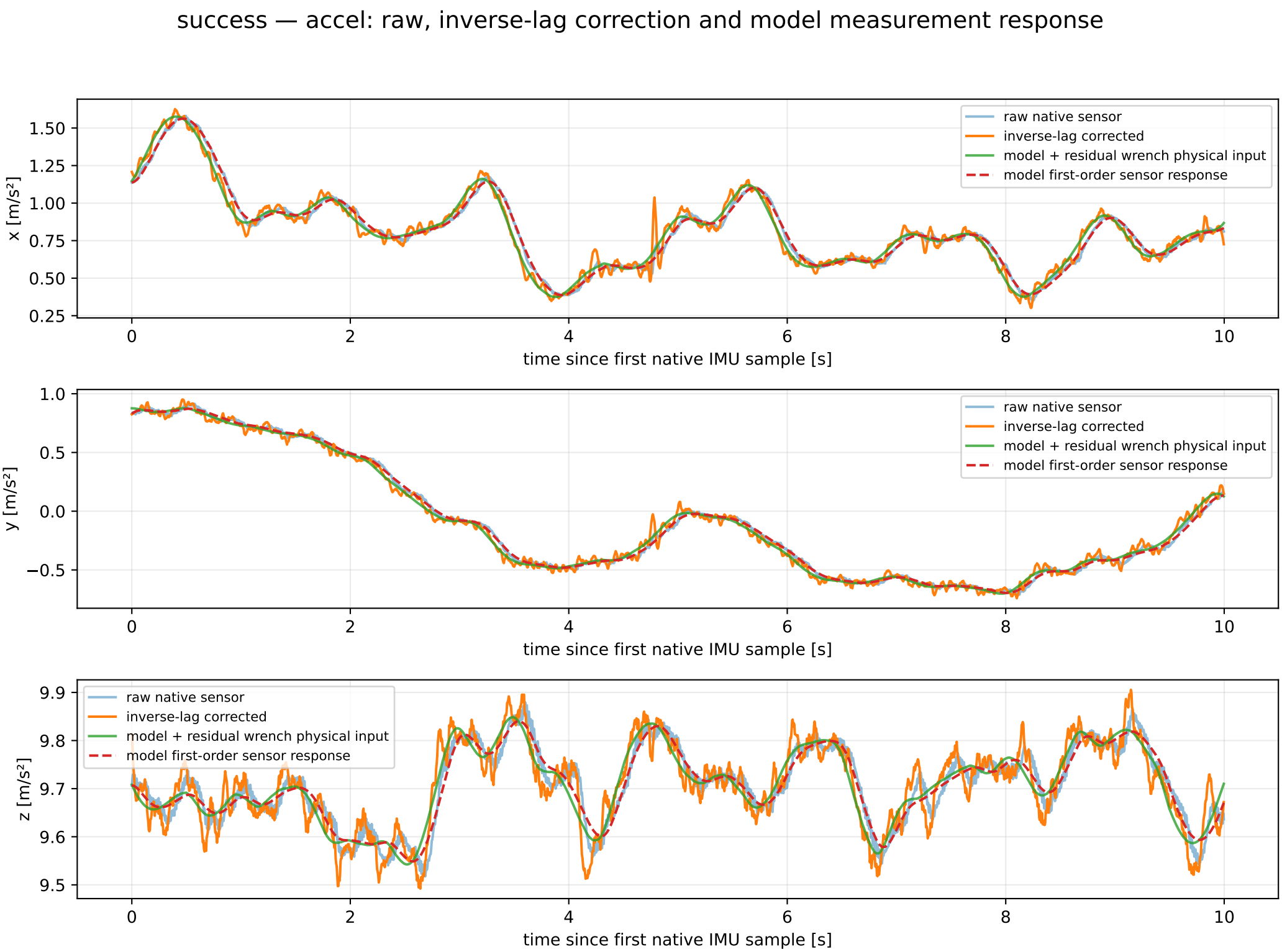

success — Generalized IMU acceleration fit
lag-corrected measurement
identified dynamics
x [m/s²]
x [rad/s²]
y [m/s²]
y [rad/s²]
z [m/s²]
z [rad/s²]
time since first native IMU sample [s]

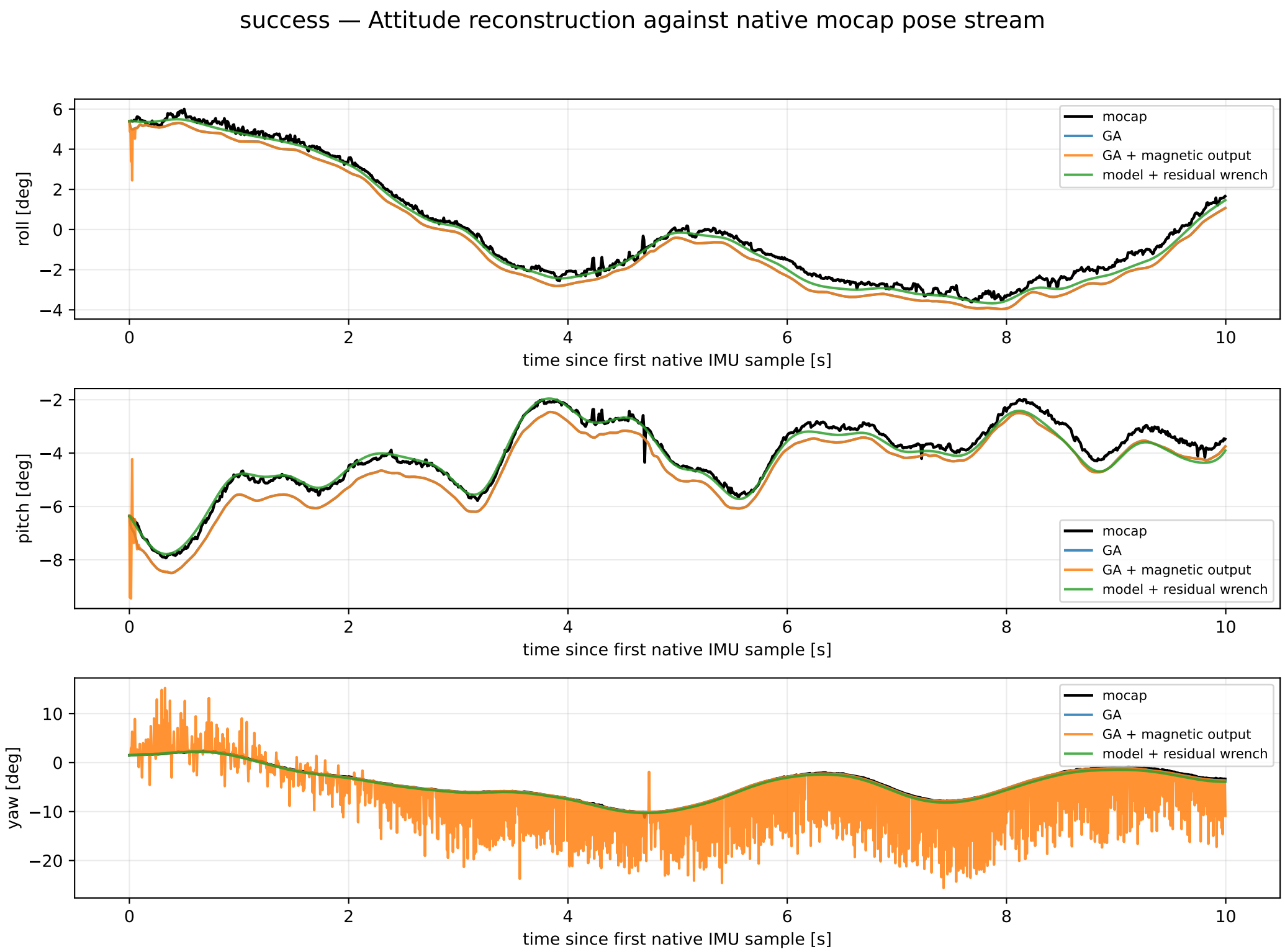

success — Residual wrench at native measurement timestamps
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
time since first native IMU sample [s]

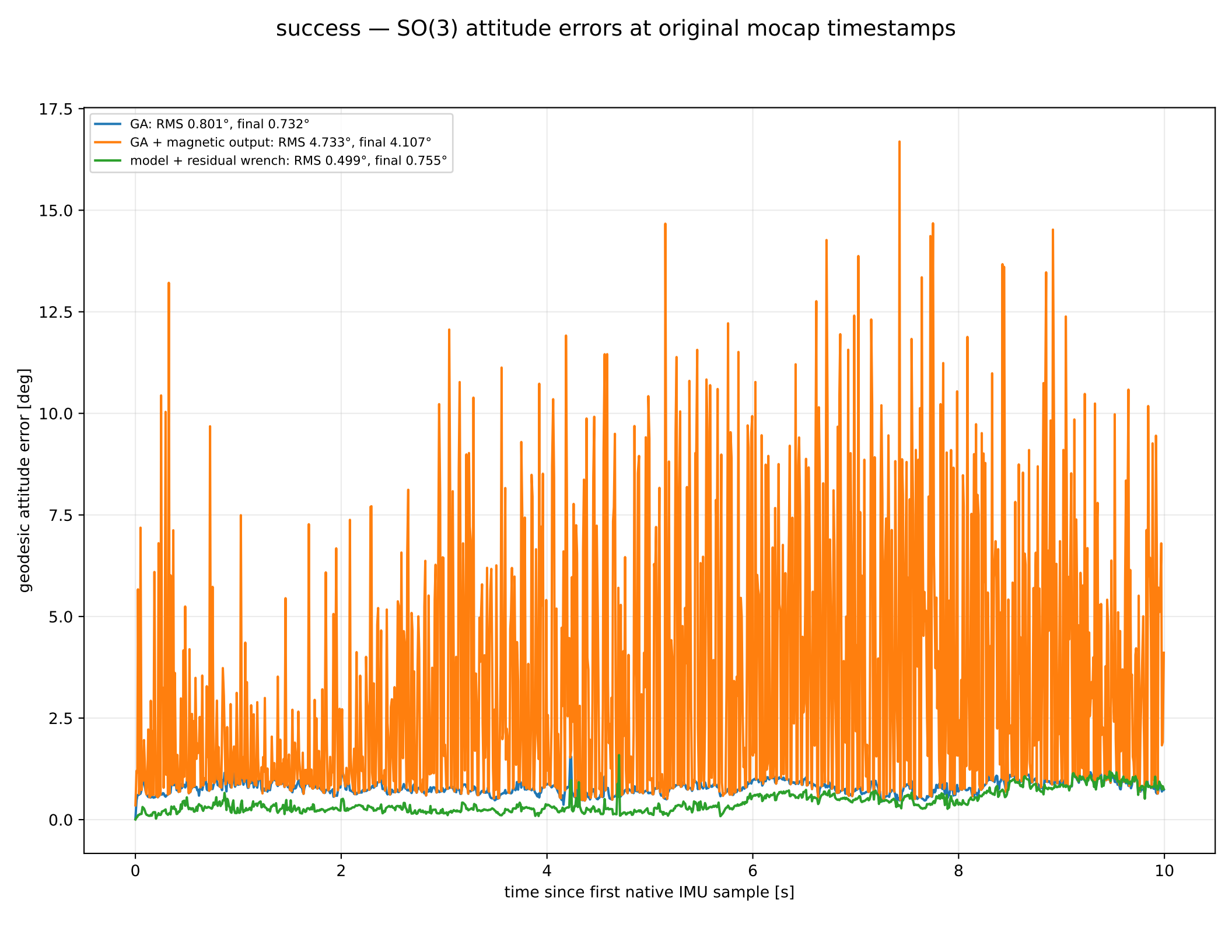

success — Residual wrench spectrum (linear detrend; native irregular clock)
Fx normalized power
Fy normalized power
Fz normalized power
Tx normalized power
Ty normalized power
Tz normalized power
frequency [Hz]

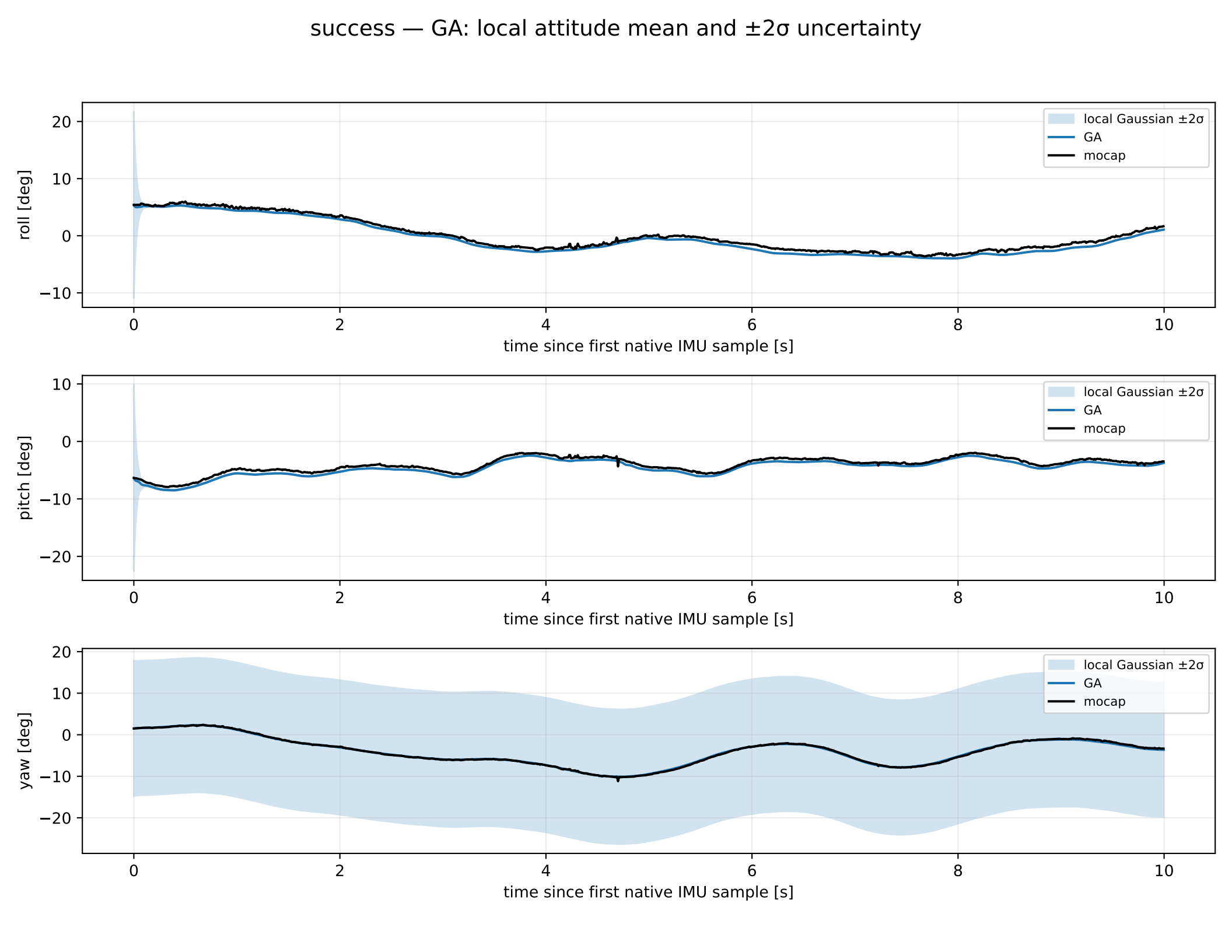

success — Applied residual wrench in reconstructed dynamics (body frame, about CoG)
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
applied law ±2σ
added residual-wrench mean
time since first native IMU sample [s]
The native-IMU residual wrench is added to the dynamics; this is not a model-only free replay.

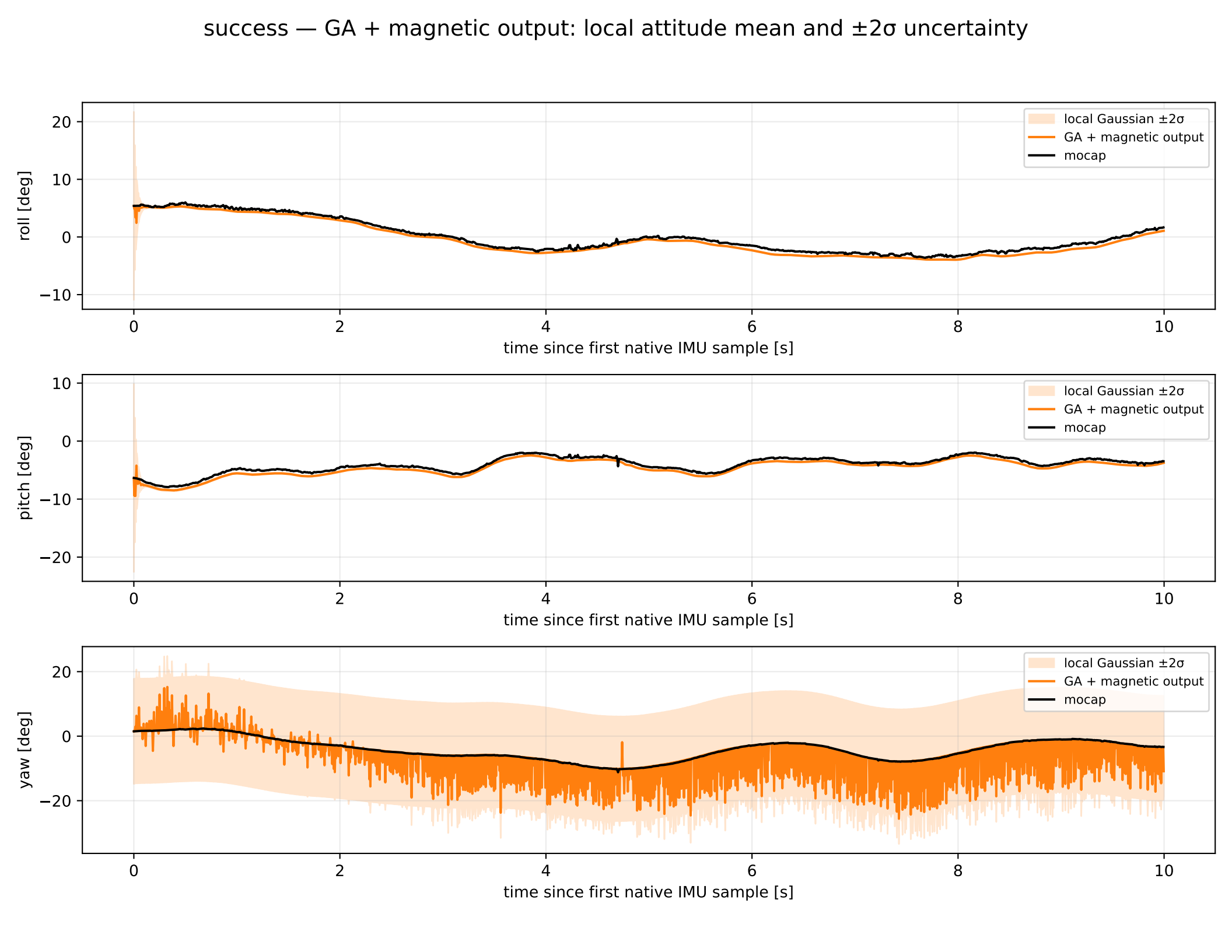

success — gyro: raw, inverse-lag correction and model measurement response
x [rad/s]
y [rad/s]
z [rad/s]
time since first native IMU sample [s]
raw native sensor
inverse-lag corrected
model + residual wrench physical input
model first-order sensor response

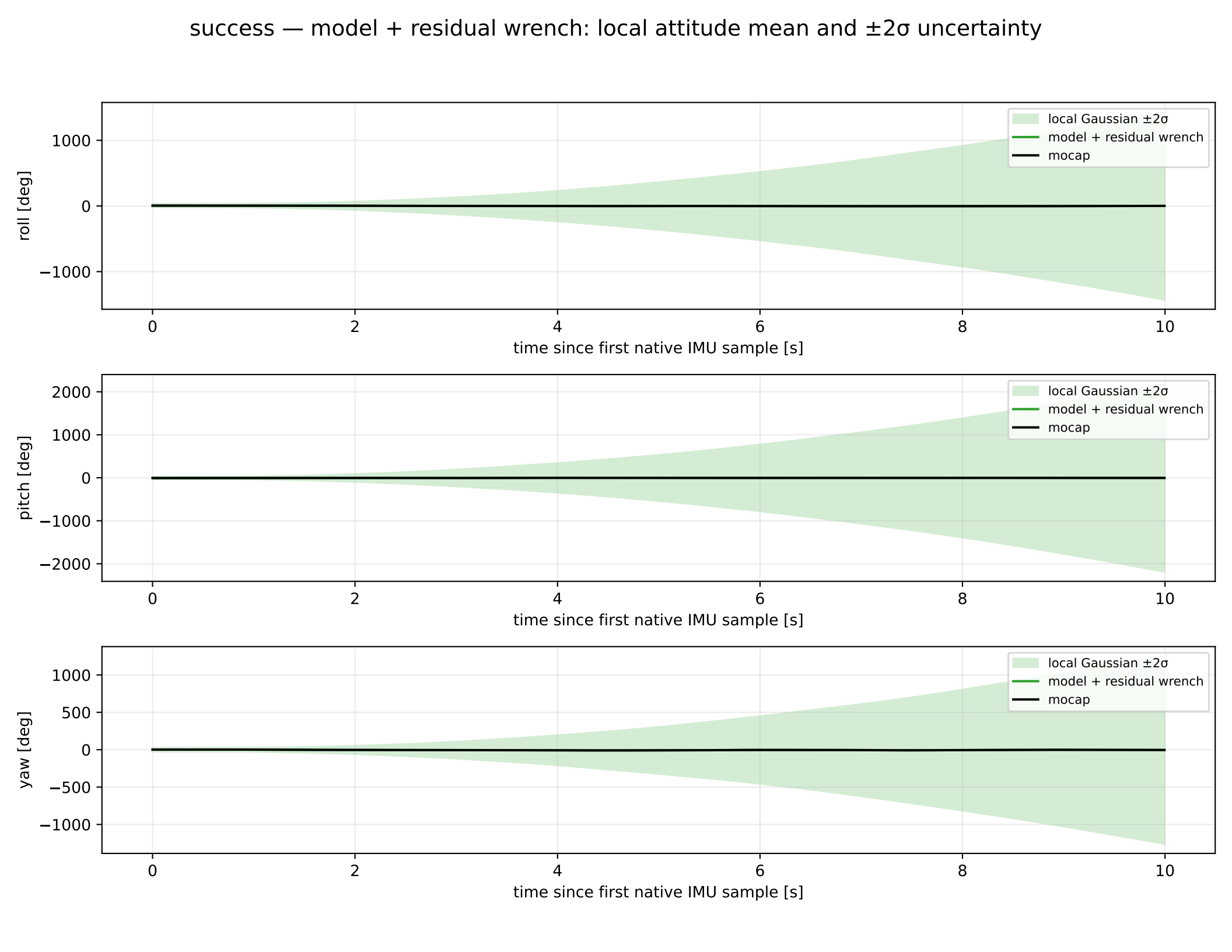

success — accel: raw, inverse-lag correction and model measurement response
raw native sensor
inverse-lag corrected
model + residual wrench physical input
model first-order sensor response
x [m/s²]
y [m/s²]
z [m/s²]
time since first native IMU sample [s]


success — Attitude reconstruction against native mocap pose stream
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
mocap
GA
GA + magnetic output
model + residual wrench

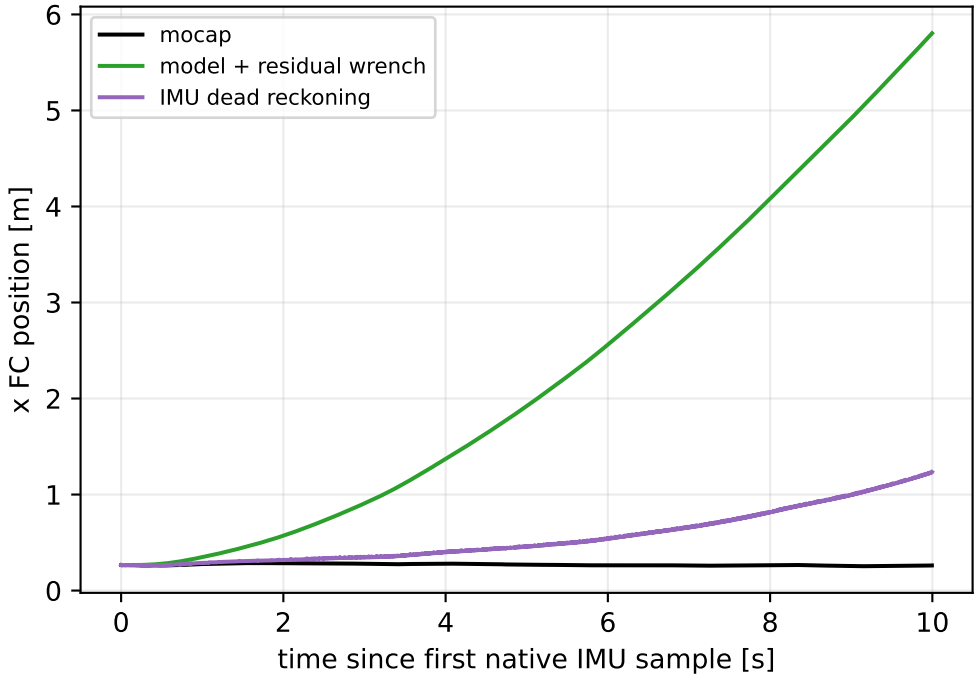

success — SO(3) attitude errors at original mocap timestamps
GA: RMS 0.801°, final 0.732°
GA + magnetic output: RMS 4.733°, final 4.107°
model + residual wrench: RMS 0.499°, final 0.755°
geodesic attitude error [deg]
time since first native IMU sample [s]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
2
4
6
8
10

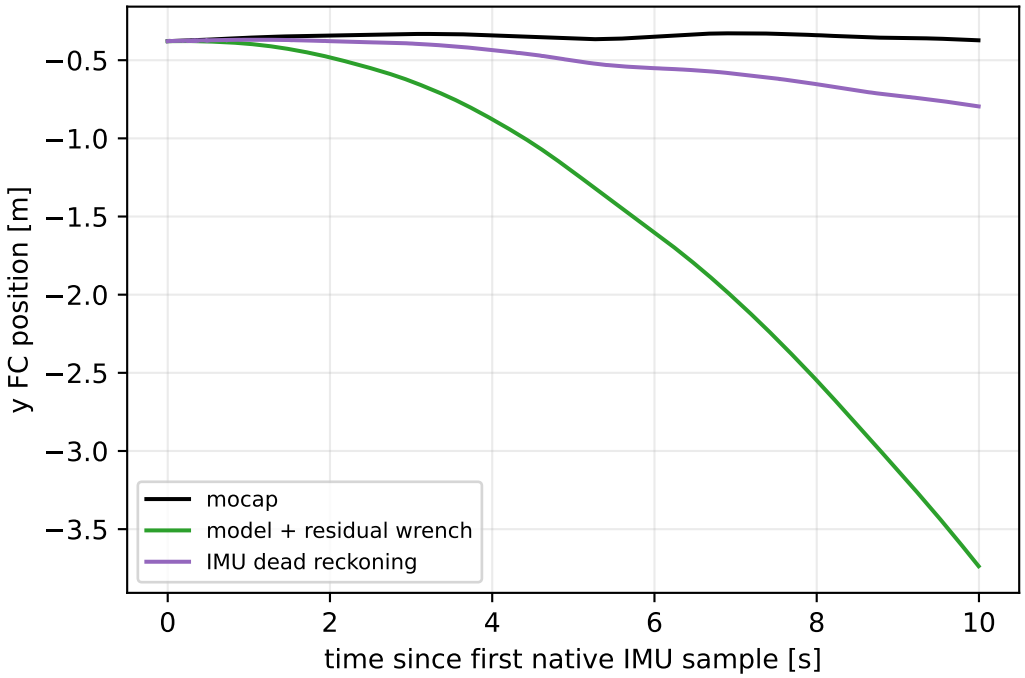

success — GA: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA
mocap

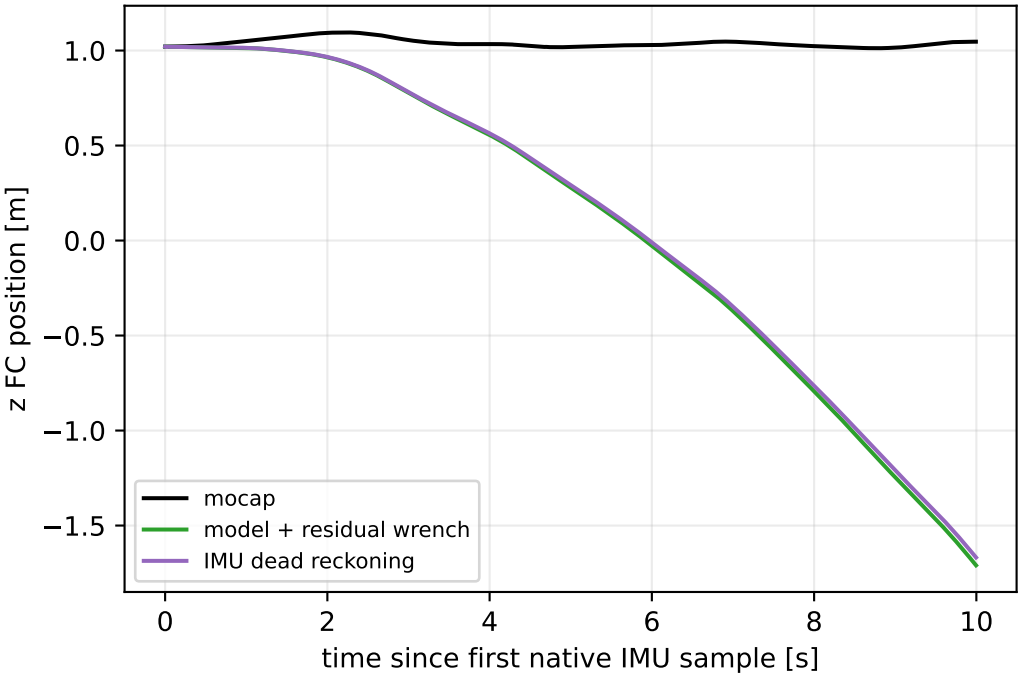

success — GA + magnetic output: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA + magnetic output
mocap

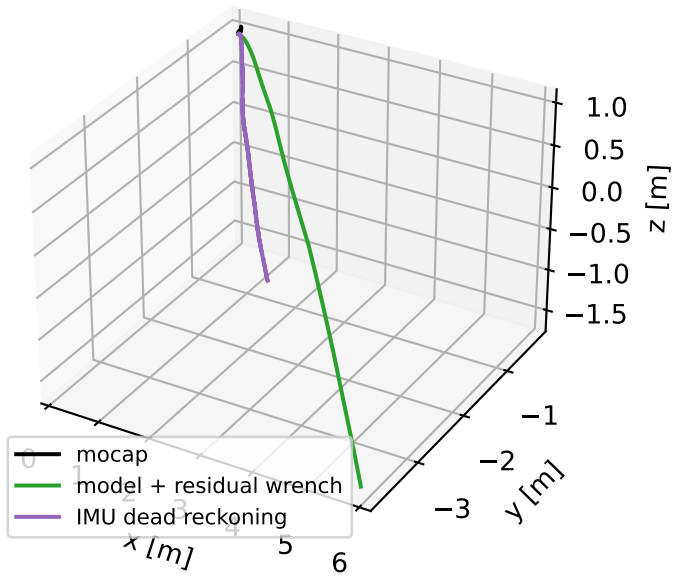

success — model + residual wrench: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
model + residual wrench
mocap

success — Position reconstruction (mean trajectories)



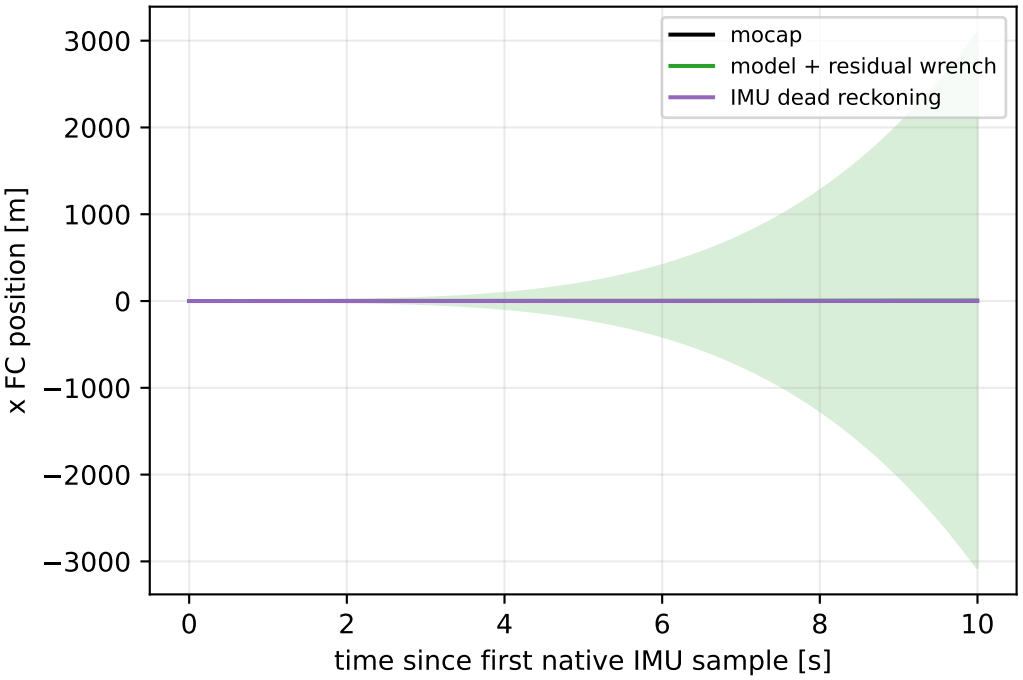


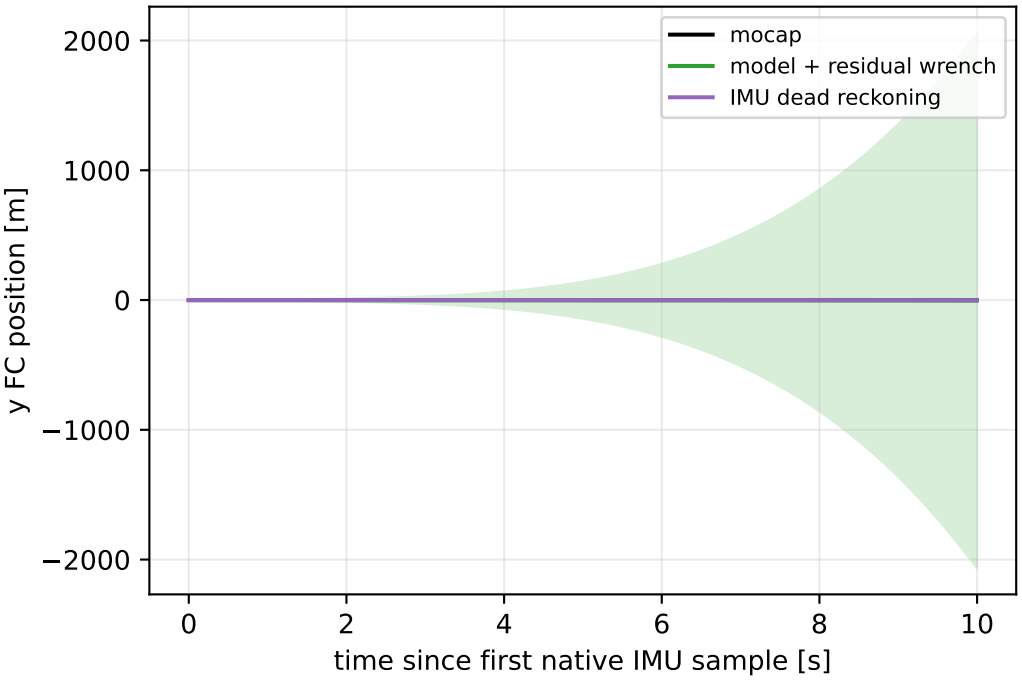


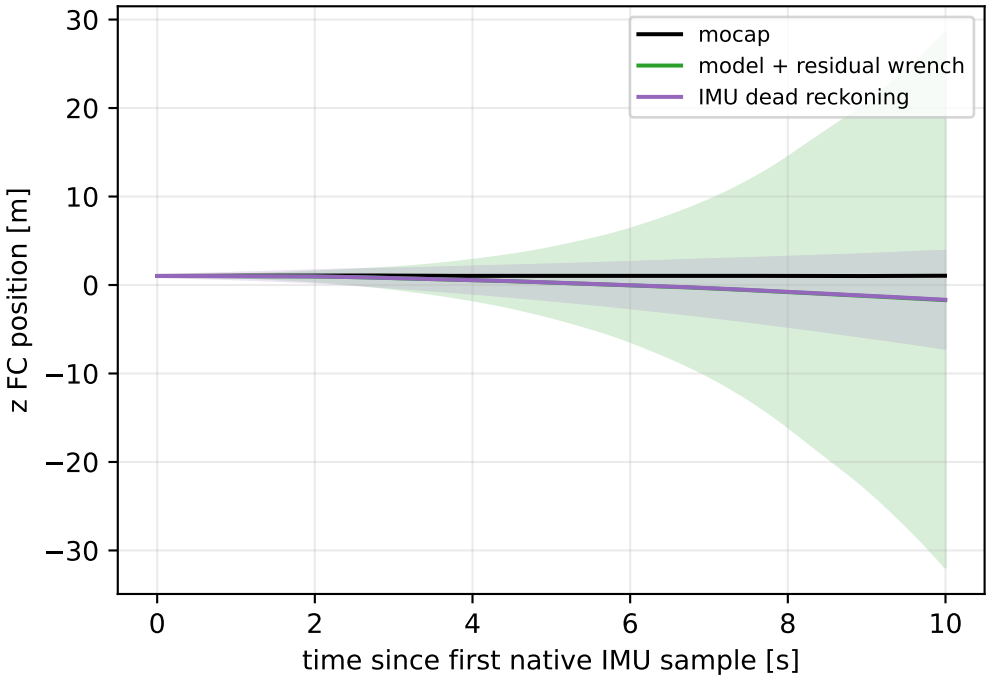


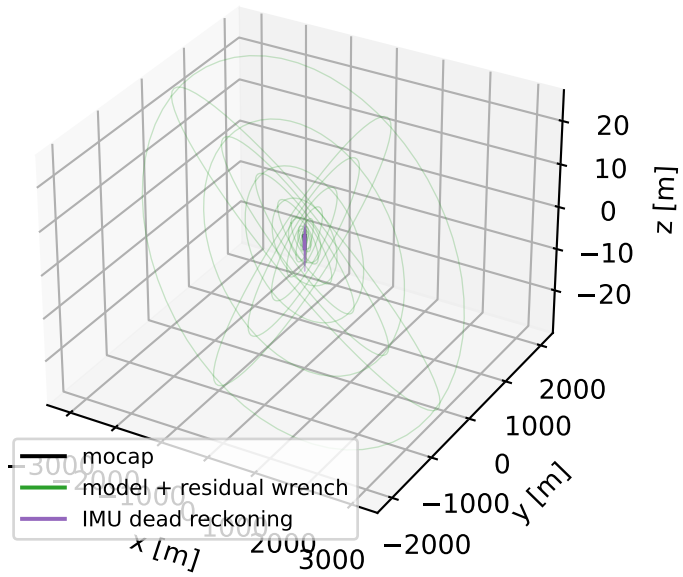
Residual-wrench model replay and IMU integration are not reset to later mocap.

success — Position reconstruction with full propagated ±2σ spread









FC origin; local Gaussian linearization. Ellipsoids: deterministic 2σ factor-SVD wireframes, not samples.

failure1 — Actual estimated SI values and joint covariance coordinate order

#	parameter	unit	estimate	std. deviation
1	mass_kg	kg	5.662356702e+00	1.930790924e+04
2	inertia_xx_kg_m2	kg m^2	3.058171573e-01	1.042792496e+03
3	inertia_yy_kg_m2	kg m^2	3.009270629e-01	1.026123399e+03
4	inertia_zz_kg_m2	kg m^2	6.044695457e-01	2.061189733e+03
5	inertia_xy_kg_m2	kg m^2	3.665243642e-03	1.249786514e+01
6	inertia_xz_kg_m2	kg m^2	-1.505029697e-03	5.134717106e+00
7	inertia_yz_kg_m2	kg m^2	1.856582079e-02	6.330401638e+01
8	cog_x_m	m	2.481261142e-02	1.549734871e-03
9	cog_y_m	m	-5.845013145e-03	9.014727386e-04
10	cog_z_m	m	-2.434450651e-02	1.074343382e-03
11	force_effectiveness_1	dimensionless	1.814096850e+00	6.185834614e+03
12	force_effectiveness_2	dimensionless	2.485234671e+00	8.474329527e+03
13	force_effectiveness_3	dimensionless	2.172043873e+00	7.406390656e+03
14	force_effectiveness_4	dimensionless	1.720953263e+00	5.868230662e+03
15	thrust_time_constant_seconds	s	2.190042293e-01	4.009869312e-03
16	gyro_time_constant_seconds	s	3.719012789e-02	1.892996386e-03
17	accel_time_constant_seconds	s	6.620588407e-02	3.042504690e-04

Common-scale gauge: physical chart projected onto existing 13-D orthogonal common-scale section; tau coordinates unaffected

Covariance units are products of row/column units. All cross terms and weak modes are retained.

Full-precision values: parameter_values_si.csv; full 17×17 covariance: parameter_covariance_si.csv.

failure1 — Full SI covariance: rows 1–9, columns 1–9

	1	2	3	4	5	6	7	8	9
1	3.727954e+08	2.013414e+07	1.981230e+07	3.979726e+07	2.413076e+05	-9.914038e+04	1.222268e+06	3.445390e-01	3.664392e+00
2	2.013414e+07	1.087416e+06	1.070034e+06	2.149393e+06	1.303268e+04	-5.354430e+03	6.601295e+04	1.861156e-02	1.979035e-01
3	1.981230e+07	1.070034e+06	1.052929e+06	2.115035e+06	1.282435e+04	-5.268839e+03	6.495773e+04	1.831100e-02	1.947444e-01
4	3.979726e+07	2.149393e+06	2.115035e+06	4.248503e+06	2.576047e+04	-1.058360e+04	1.304816e+05	3.678449e-02	3.911820e-01
5	2.413076e+05	1.303268e+04	1.282435e+04	2.576047e+04	1.561966e+02	-6.417284e+01	7.911651e+02	2.224373e-04	2.372069e-03
6	-9.914038e+04	-5.354430e+03	-5.268839e+03	-1.058360e+04	-6.417284e+01	2.636532e+01	-3.250473e+02	-8.757603e-05	-9.765488e-04
7	1.222268e+06	6.601295e+04	6.495773e+04	1.304816e+05	7.911651e+02	-3.250473e+02	4.007398e+03	1.130850e-03	1.201249e-02
8	3.445390e-01	1.861156e-02	1.831100e-02	3.678449e-02	2.224373e-04	-8.757603e-05	1.130850e-03	2.401678e-06	-4.419781e-07
9	3.664392e+00	1.979035e-01	1.947444e-01	3.911820e-01	2.372069e-03	-9.765488e-04	1.201249e-02	-4.419781e-07	8.126531e-07

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure1 — Full SI covariance: rows 1–9, columns 10–17

	10	11	12	13	14	15	16	17
1	-3.212669e+00	1.194355e+08	1.636216e+08	1.430019e+08	1.133033e+08	-7.492179e+00	6.046955e+00	1.781598e-01
2	-1.735176e-01	6.450542e+06	8.836967e+06	7.723329e+06	6.119347e+06	-4.046489e-01	3.265867e-01	9.622524e-03
3	-1.707430e-01	6.347430e+06	8.695708e+06	7.599871e+06	6.021529e+06	-3.981794e-01	3.213651e-01	9.468692e-03
4	-3.429751e-01	1.275018e+07	1.746720e+07	1.526598e+07	1.209554e+07	-7.998307e-01	6.455322e-01	1.901997e-02
5	-2.079012e-03	7.730973e+04	1.059110e+05	9.256407e+04	7.334036e+04	-4.849407e-03	3.914340e-03	1.152546e-04
6	8.540140e-04	-3.176243e+04	-4.351317e+04	-3.802962e+04	-3.013162e+04	2.002309e-03	-1.609694e-03	-4.723286e-05
7	-1.053325e-02	3.915882e+05	5.364591e+05	4.688543e+05	3.714826e+05	-2.456180e-02	1.982510e-02	5.839530e-04
8	-1.367391e-07	1.103637e-01	1.512479e-01	1.321798e-01	1.046921e-01	-1.749291e-06	2.582529e-07	9.958268e-08
9	1.834891e-07	1.173989e+00	1.608306e+00	1.405639e+00	1.113724e+00	-6.724622e-07	5.948103e-07	4.787001e-08

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure1 — Full SI covariance: rows 10–17, columns 1–9

	1	2	3	4	5	6	7	8	9
10	-3.212669e+00	-1.735176e-01	-1.707430e-01	-3.429751e-01	-2.079012e-03	8.540140e-04	-1.053325e-02	-1.367391e-07	1.834891e-07
11	1.194355e+08	6.450542e+06	6.347430e+06	1.275018e+07	7.730973e+04	-3.176243e+04	3.915882e+05	1.103637e-01	1.173989e+00
12	1.636216e+08	8.836967e+06	8.695708e+06	1.746720e+07	1.059110e+05	-4.351317e+04	5.364591e+05	1.512479e-01	1.608306e+00
13	1.430019e+08	7.723329e+06	7.599871e+06	1.526598e+07	9.256407e+04	-3.802962e+04	4.688543e+05	1.321798e-01	1.405639e+00
14	1.133033e+08	6.119347e+06	6.021529e+06	1.209554e+07	7.334036e+04	-3.013162e+04	3.714826e+05	1.046921e-01	1.113724e+00
15	-7.492179e+00	-4.046489e-01	-3.981794e-01	-7.998307e-01	-4.849407e-03	2.002309e-03	-2.456180e-02	-1.749291e-06	-6.724622e-07
16	6.046955e+00	3.265867e-01	3.213651e-01	6.455322e-01	3.914340e-03	-1.609694e-03	1.982510e-02	2.582529e-07	5.948103e-07
17	1.781598e-01	9.622524e-03	9.468692e-03	1.901997e-02	1.152546e-04	-4.723286e-05	5.839530e-04	9.958268e-08	4.787001e-08

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure1 — Full SI covariance: rows 10–17, columns 10–17

	10	11	12	13	14	15	16	17
10	1.154214e-06	-1.029271e+00	-1.410060e+00	-1.232360e+00	-9.764187e-01	4.233956e-07	6.557444e-07	-3.506045e-08
11	-1.029271e+00	3.826455e+07	5.242080e+07	4.581471e+07	3.629990e+07	-2.400313e+00	1.937305e+00	5.707725e-02
12	-1.410060e+00	5.242080e+07	7.181426e+07	6.276420e+07	4.972932e+07	-3.288367e+00	2.654033e+00	7.819603e-02
13	-1.232360e+00	4.581471e+07	6.276420e+07	5.485462e+07	4.346241e+07	-2.873972e+00	2.319578e+00	6.834223e-02
14	-9.764187e-01	3.629990e+07	4.972932e+07	4.346241e+07	3.443613e+07	-2.277082e+00	1.837848e+00	5.414755e-02
15	4.233956e-07	-2.400313e+00	-3.288367e+00	-2.873972e+00	-2.277082e+00	1.607905e-05	-5.686609e-06	-7.164284e-07
16	6.557444e-07	1.937305e+00	2.654033e+00	2.319578e+00	1.837848e+00	-5.686609e-06	3.583435e-06	2.708871e-07
17	-3.506045e-08	5.707725e-02	7.819603e-02	6.834223e-02	5.414755e-02	-7.164284e-07	2.708871e-07	9.256835e-08

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure2 — Actual estimated SI values and joint covariance coordinate order

#	parameter	unit	estimate	std. deviation
1	mass_kg	kg	4.202190638e+00	2.120401021e+03
2	inertia_xx_kg_m2	kg m^2	3.251766764e-01	1.640699217e+02
3	inertia_yy_kg_m2	kg m^2	9.977341905e-02	5.034457425e+01
4	inertia_zz_kg_m2	kg m^2	3.814040225e-01	1.924906717e+02
5	inertia_xy_kg_m2	kg m^2	5.313074270e-03	2.679818991e+00
6	inertia_xz_kg_m2	kg m^2	3.438717982e-02	1.737091680e+01
7	inertia_yz_kg_m2	kg m^2	-4.707781661e-02	2.375683404e+01
8	cog_x_m	m	-4.500179327e-03	2.240183553e-03
9	cog_y_m	m	8.127700298e-03	9.877383018e-04
10	cog_z_m	m	3.331267616e-03	1.050935365e-03
11	force_effectiveness_1	dimensionless	1.521768316e+00	7.678800039e+02
12	force_effectiveness_2	dimensionless	1.412146048e+00	7.125534991e+02
13	force_effectiveness_3	dimensionless	1.607095682e+00	8.109241551e+02
14	force_effectiveness_4	dimensionless	1.487139969e+00	7.504091076e+02
15	thrust_time_constant_seconds	s	3.552099499e-01	6.526905702e-03
16	gyro_time_constant_seconds	s	6.159870505e-02	3.906426236e-03
17	accel_time_constant_seconds	s	6.209106763e-02	8.514447286e-04

Common-scale gauge: physical chart projected onto existing 13-D orthogonal common-scale section; tau coordinates unaffected

Covariance units are products of row/column units. All cross terms and weak modes are retained.

Full-precision values: parameter_values_si.csv; full 17×17 covariance: parameter_covariance_si.csv.

failure2 — Full SI covariance: rows 1–9, columns 1–9

	1	2	3	4	5	6	7	8	9
1	4.496100e+06	3.478940e+05	1.067507e+05	4.081574e+05	5.682290e+03	3.683329e+04	-5.037401e+04	-1.984904e+00	1.288606e-01
2	3.478940e+05	2.691894e+04	8.260030e+03	3.158193e+04	4.396776e+02	2.850044e+03	-3.897782e+03	-1.536145e-01	9.986693e-03
3	1.067507e+05	8.260030e+03	2.534576e+03	9.690861e+03	1.349143e+02	8.745310e+02	-1.196028e+03	-4.713276e-02	3.062253e-03
4	4.081574e+05	3.158193e+04	9.690861e+03	3.705266e+04	5.158401e+02	3.343738e+03	-4.572969e+03	-1.801826e-01	1.169282e-02
5	5.682290e+03	4.396776e+02	1.349143e+02	5.158401e+02	7.181430e+00	4.655089e+01	-6.366400e+01	-2.509028e-03	1.629895e-04
6	3.683329e+04	2.850044e+03	8.745310e+02	3.343738e+03	4.655089e+01	3.017488e+02	-4.126778e+02	-1.628059e-02	1.066597e-03
7	-5.037401e+04	-3.897782e+03	-1.196028e+03	-4.572969e+03	-6.366400e+01	-4.126778e+02	5.643872e+02	2.224296e-02	-1.446684e-03
8	-1.984904e+00	-1.536145e-01	-4.713276e-02	-1.801826e-01	-2.509028e-03	-1.628059e-02	2.224296e-02	5.018422e-06	-1.564380e-06
9	1.288606e-01	9.986693e-03	3.062253e-03	1.169282e-02	1.629895e-04	1.066597e-03	-1.446684e-03	-1.564380e-06	9.756270e-07

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure2 — Full SI covariance: rows 1–9, columns 10–17

	10	11	12	13	14	15	16	17
1	2.220203e-01	1.628214e+06	1.510899e+06	1.719484e+06	1.591168e+06	9.598091e-01	1.770249e+00	-7.991522e-01
2	1.716198e-02	1.259860e+05	1.169086e+05	1.330483e+05	1.231196e+05	7.424657e-02	1.369361e-01	-6.183330e-02
3	5.266943e-03	3.865859e+04	3.587320e+04	4.082563e+04	3.777903e+04	2.277687e-02	4.202278e-02	-1.897410e-02
4	2.015259e-02	1.478097e+05	1.371599e+05	1.560953e+05	1.444468e+05	8.715126e-02	1.606854e-01	-7.254145e-02
5	2.803725e-04	2.057779e+03	1.909514e+03	2.173130e+03	2.010960e+03	1.210109e-03	2.238112e-03	-1.010058e-03
6	1.809200e-03	1.333877e+04	1.237770e+04	1.408649e+04	1.303529e+04	7.835290e-03	1.448805e-02	-6.547959e-03
7	-2.485659e-03	-1.824240e+04	-1.692801e+04	-1.926499e+04	-1.782734e+04	-1.075491e-02	-1.982759e-02	8.954450e-03
8	1.082730e-06	-7.188254e-01	-6.669849e-01	-7.590842e-01	-7.024953e-01	2.916406e-06	1.236765e-06	8.418126e-07
9	-6.008383e-07	4.666920e-02	4.328644e-02	4.927784e-02	4.561924e-02	-8.947103e-07	-1.231535e-06	-2.808812e-07

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure2 — Full SI covariance: rows 10–17, columns 1–9

	1	2	3	4	5	6	7	8	9
10	2.220203e-01	1.716198e-02	5.266943e-03	2.015259e-02	2.803725e-04	1.809200e-03	-2.485659e-03	1.082730e-06	-6.008383e-07
11	1.628214e+06	1.259860e+05	3.865859e+04	1.478097e+05	2.057779e+03	1.333877e+04	-1.824240e+04	-7.188254e-01	4.666920e-02
12	1.510899e+06	1.169086e+05	3.587320e+04	1.371599e+05	1.909514e+03	1.237770e+04	-1.692801e+04	-6.669849e-01	4.328644e-02
13	1.719484e+06	1.330483e+05	4.082563e+04	1.560953e+05	2.173130e+03	1.408649e+04	-1.926499e+04	-7.590842e-01	4.927784e-02
14	1.591168e+06	1.231196e+05	3.777903e+04	1.444468e+05	2.010960e+03	1.303529e+04	-1.782734e+04	-7.024953e-01	4.561924e-02
15	9.598091e-01	7.424657e-02	2.277687e-02	8.715126e-02	1.210109e-03	7.835290e-03	-1.075491e-02	2.916406e-06	-8.947103e-07
16	1.770249e+00	1.369361e-01	4.202278e-02	1.606854e-01	2.238112e-03	1.448805e-02	-1.982759e-02	1.236765e-06	-1.231535e-06
17	-7.991522e-01	-6.183330e-02	-1.897410e-02	-7.254145e-02	-1.010058e-03	-6.547959e-03	8.954450e-03	8.418126e-07	-2.808812e-07

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure2 — Full SI covariance: rows 10–17, columns 10–17

	10	11	12	13	14	15	16	17
10	1.104465e-06	8.039873e-02	7.462070e-02	8.491301e-02	7.856211e-02	1.369631e-06	2.631578e-06	-6.290448e-08
11	8.039873e-02	5.896397e+05	5.471556e+05	6.226924e+05	5.762241e+05	3.475666e-01	6.410718e-01	-2.894063e-01
12	7.462070e-02	5.471556e+05	5.077325e+05	5.778268e+05	5.347066e+05	3.225737e-01	5.949057e-01	-2.685470e-01
13	8.491301e-02	6.226924e+05	5.778268e+05	6.575980e+05	6.085249e+05	3.670796e-01	6.770175e-01	-3.056261e-01
14	7.856211e-02	5.762241e+05	5.347066e+05	6.085249e+05	5.631138e+05	3.396533e-01	6.264697e-01	-2.828245e-01
15	1.369631e-06	3.475666e-01	3.225737e-01	3.670796e-01	3.396533e-01	4.260050e-05	-1.272511e-05	-2.385254e-07
16	2.631578e-06	6.410718e-01	5.949057e-01	6.770175e-01	6.264697e-01	-1.272511e-05	1.526017e-05	3.588101e-08
17	-6.290448e-08	-2.894063e-01	-2.685470e-01	-3.056261e-01	-2.828245e-01	-2.385254e-07	3.588101e-08	7.249581e-07

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

success — Actual estimated SI values and joint covariance coordinate order

#	parameter	unit	estimate	std. deviation
1	mass_kg	kg	2.858943911e+00	2.595899936e+02
2	inertia_xx_kg_m2	kg m^2	5.538736339e-01	5.032196978e+01
3	inertia_yy_kg_m2	kg m^2	2.269874855e-01	2.061466211e+01
4	inertia_zz_kg_m2	kg m^2	3.570246313e-01	3.241100546e+01
5	inertia_xy_kg_m2	kg m^2	-5.996776904e-02	5.449286735e+00
6	inertia_xz_kg_m2	kg m^2	2.349699796e-02	2.141092361e+00
7	inertia_yz_kg_m2	kg m^2	-4.059893266e-02	3.683411569e+00
8	cog_x_m	m	-1.459485598e-02	1.290555178e-03
9	cog_y_m	m	-3.770966559e-02	6.519965024e-04
10	cog_z_m	m	-7.308516943e-04	2.753033284e-03
11	force_effectiveness_1	dimensionless	1.257231281e+00	1.141591796e+02
12	force_effectiveness_2	dimensionless	1.077551658e+00	9.783795813e+01
13	force_effectiveness_3	dimensionless	8.497428069e-01	7.715238664e+01
14	force_effectiveness_4	dimensionless	7.583319663e-01	6.885846628e+01
15	thrust_time_constant_seconds	s	1.977120296e-01	1.197364424e-02
16	gyro_time_constant_seconds	s	3.093824562e-02	2.212413488e-03
17	accel_time_constant_seconds	s	6.318998861e-02	2.094005976e-03

Common-scale gauge: physical chart projected onto existing 13-D orthogonal common-scale section; tau coordinates unaffected

Covariance units are products of row/column units. All cross terms and weak modes are retained.

Full-precision values: parameter_values_si.csv; full 17×17 covariance: parameter_covariance_si.csv.

success — Full SI covariance: rows 1–9, columns 1–9

	1	2	3	4	5	6	7	8	9
1	6.738696e+04	1.306308e+04	5.351360e+03	8.413572e+03	-1.414579e+03	5.557988e+02	-9.561753e+02	-2.291420e-01	-2.213826e-02
2	1.306308e+04	2.532301e+03	1.037370e+03	1.630986e+03	-2.742186e+02	1.077425e+02	-1.853563e+02	-4.442088e-02	-4.289122e-03
3	5.351360e+03	1.037370e+03	4.249643e+02	6.681419e+02	-1.123351e+02	4.413730e+01	-7.593219e+01	-1.819796e-02	-1.757338e-03
4	8.413572e+03	1.630986e+03	6.681419e+02	1.050473e+03	-1.766167e+02	6.939409e+01	-1.193829e+02	-2.861258e-02	-2.760392e-03
5	-1.414579e+03	-2.742186e+02	-1.123351e+02	-1.766167e+02	2.969473e+01	-1.166730e+01	2.007193e+01	4.810466e-03	4.636083e-04
6	5.557988e+02	1.077425e+02	4.413730e+01	6.939409e+01	-1.166730e+01	4.584276e+00	-7.886433e+00	-1.894464e-03	-1.808476e-04
7	-9.561753e+02	-1.853563e+02	-7.593219e+01	-1.193829e+02	2.007193e+01	-7.886433e+00	1.356752e+01	3.253486e-03	3.118245e-04
8	-2.291420e-01	-4.442088e-02	-1.819796e-02	-2.861258e-02	4.810466e-03	-1.894464e-03	3.253486e-03	1.665533e-06	2.377747e-09
9	-2.213826e-02	-4.289122e-03	-1.757338e-03	-2.760392e-03	4.636083e-04	-1.808476e-04	3.118245e-04	2.377747e-09	4.250994e-07

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

success — Full SI covariance: rows 1–9, columns 10–17

	10	11	12	13	14	15	16	17
1	-5.646762e-01	2.963458e+04	2.539775e+04	2.002799e+04	1.787497e+04	-1.463135e+00	-2.417474e-04	1.889011e-01
2	-1.094826e-01	5.744714e+03	4.923398e+03	3.882460e+03	3.465093e+03	-2.837299e-01	-4.946739e-05	3.662433e-02
3	-4.485310e-02	2.353353e+03	2.016896e+03	1.590470e+03	1.419494e+03	-1.162578e-01	-1.730520e-05	1.500415e-02
4	-7.051110e-02	3.700014e+03	3.171026e+03	2.500586e+03	2.231772e+03	-1.826853e-01	-3.242878e-05	2.358563e-02
5	1.185420e-02	-6.220856e+02	-5.331466e+02	-4.204251e+02	-3.752292e+02	3.069882e-02	4.225873e-06	-3.964057e-03
6	-4.658286e-03	2.444221e+02	2.094773e+02	1.651882e+02	1.474304e+02	-1.205187e-02	6.608024e-07	1.556730e-03
7	8.016775e-03	-4.204946e+02	-3.603769e+02	-2.841836e+02	-2.536337e+02	2.076267e-02	5.107052e-06	-2.681077e-03
8	2.577171e-06	-1.007728e-01	-8.635759e-02	-6.809979e-02	-6.078524e-02	5.341716e-06	-9.078042e-08	-8.204725e-07
9	-1.955527e-07	-9.736762e-03	-8.346371e-03	-6.577755e-03	-5.870856e-03	9.432724e-07	-3.621833e-08	6.280862e-08

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

success — Full SI covariance: rows 10–17, columns 1–9

	1	2	3	4	5	6	7	8	9
10	-5.646762e-01	-1.094826e-01	-4.485310e-02	-7.051110e-02	1.185420e-02	-4.658286e-03	8.016775e-03	2.577171e-06	-1.955527e-07
11	2.963458e+04	5.744714e+03	2.353353e+03	3.700014e+03	-6.220856e+02	2.444221e+02	-4.204946e+02	-1.007728e-01	-9.736762e-03
12	2.539775e+04	4.923398e+03	2.016896e+03	3.171026e+03	-5.331466e+02	2.094773e+02	-3.603769e+02	-8.635759e-02	-8.346371e-03
13	2.002799e+04	3.882460e+03	1.590470e+03	2.500586e+03	-4.204251e+02	1.651882e+02	-2.841836e+02	-6.809979e-02	-6.577755e-03
14	1.787497e+04	3.465093e+03	1.419494e+03	2.231772e+03	-3.752292e+02	1.474304e+02	-2.536337e+02	-6.078524e-02	-5.870856e-03
15	-1.463135e+00	-2.837299e-01	-1.162578e-01	-1.826853e-01	3.069882e-02	-1.205187e-02	2.076267e-02	5.341716e-06	9.432724e-07
16	-2.417474e-04	-4.946739e-05	-1.730520e-05	-3.242878e-05	4.225873e-06	6.608024e-07	5.107052e-06	-9.078042e-08	-3.621833e-08
17	1.889011e-01	3.662433e-02	1.500415e-02	2.358563e-02	-3.964057e-03	1.556730e-03	-2.681077e-03	-8.204725e-07	6.280862e-08

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

success — Full SI covariance: rows 10–17, columns 10–17

	10	11	12	13	14	15	16	17
10	7.579192e-06	-2.483271e-01	-2.128187e-01	-1.678269e-01	-1.497879e-01	2.229369e-05	-1.474582e-07	-2.301668e-06
11	-2.483271e-01	1.303232e+04	1.116910e+04	8.807653e+03	7.860826e+03	-6.434386e-01	-1.055746e-04	8.307276e-02
12	-2.128187e-01	1.116910e+04	9.572266e+03	7.548432e+03	6.736972e+03	-5.514531e-01	-9.153034e-05	7.119410e-02
13	-1.678269e-01	8.807653e+03	7.548432e+03	5.952491e+03	5.312595e+03	-4.348544e-01	-7.222006e-05	5.614351e-02
14	-1.497879e-01	7.860826e+03	6.736972e+03	5.312595e+03	4.741488e+03	-3.881062e-01	-6.415081e-05	5.010837e-02
15	2.229369e-05	-6.434386e-01	-5.514531e-01	-4.348544e-01	-3.881062e-01	1.433682e-04	-1.113262e-05	-7.428792e-06
16	-1.474582e-07	-1.055746e-04	-9.153034e-05	-7.222006e-05	-6.415081e-05	-1.113262e-05	4.894773e-06	6.262505e-07
17	-2.301668e-06	8.307276e-02	7.119410e-02	5.614351e-02	5.010837e-02	-7.428792e-06	6.262505e-07	4.384861e-06

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.